Google AI Classroom

Architecting Edge-to-Cloud Multimodal AI with Google Cloud, Firebase & Gemini

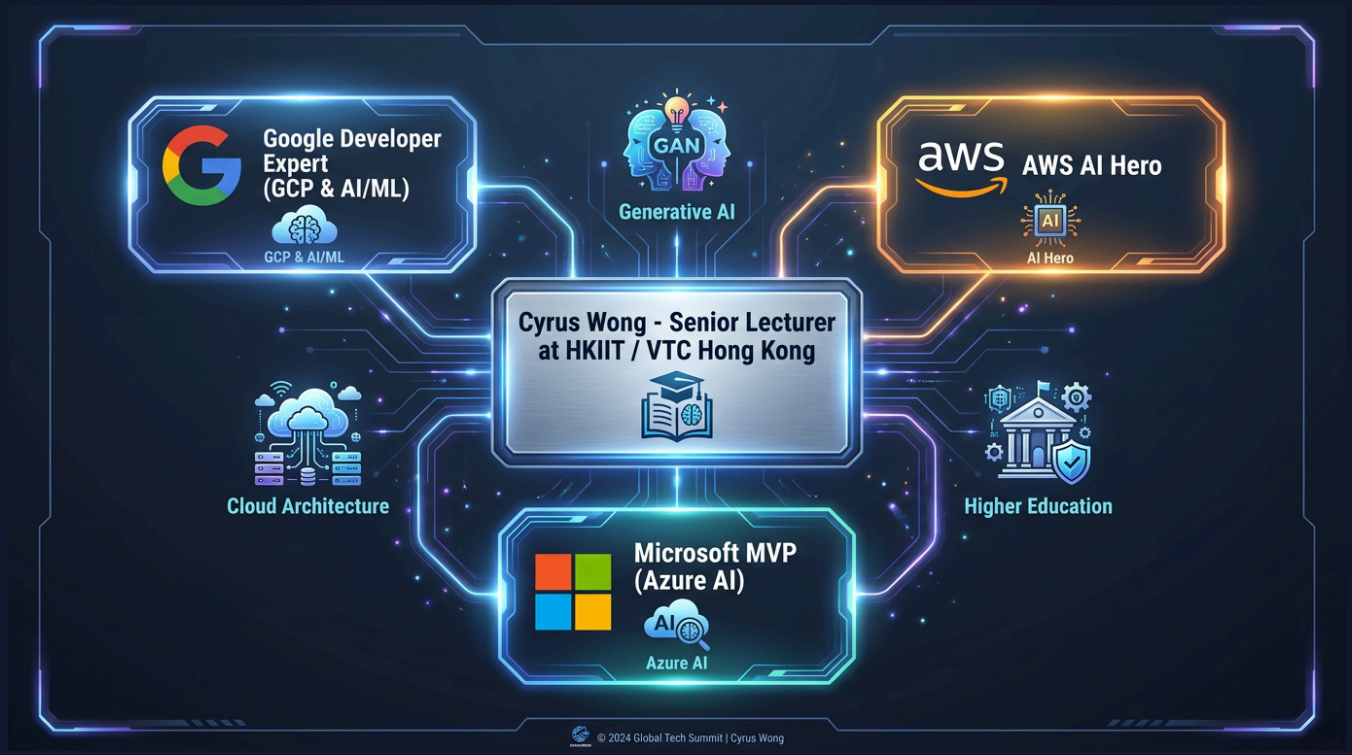
Google Cloud Tech Talk & Developer Conference Series

Presenter: Cyrus Wong (黃俊彥) — Google Developer Expert (GCP & AI/ML)
Senior Lecturer, Hong Kong Institute of Information Technology (HKIIT), VTC Hong Kong

About the Speaker: Cyrus Wong (黃俊彥)

Senior Lecturer, HKIIT / VTC Hong Kong & Google Developer Expert

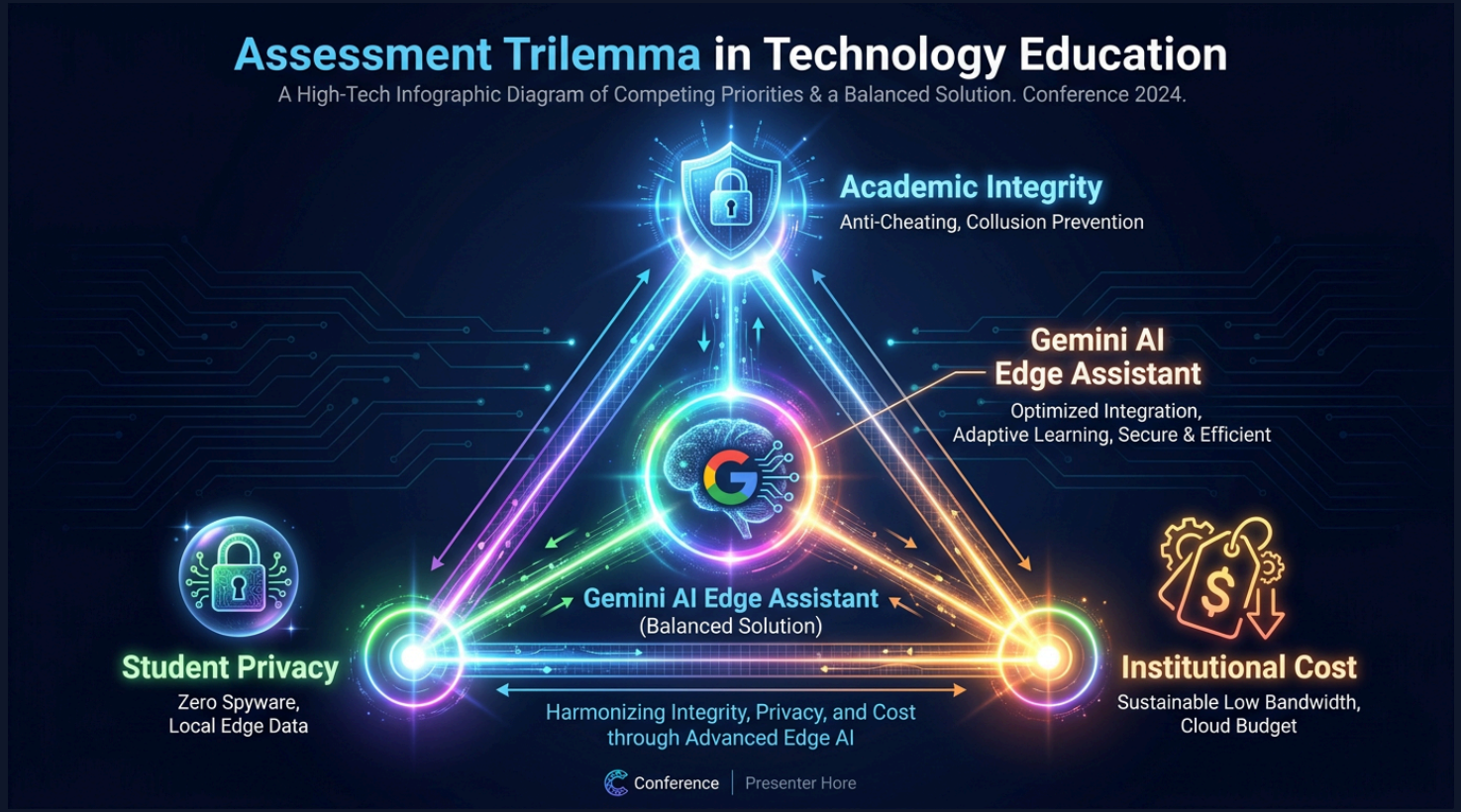
- **Institution:** Hong Kong Institute of Information Technology (HKIIT), Vocational Training Council (VTC)
- **Academic Focus:** Cloud & Data Centre Administration, Edge AI Systems Architecture



01 | The Real-Time Invigilation & Assessment Challenge

Why Traditional Proctoring and Commercial Surveillance Fail at Scale

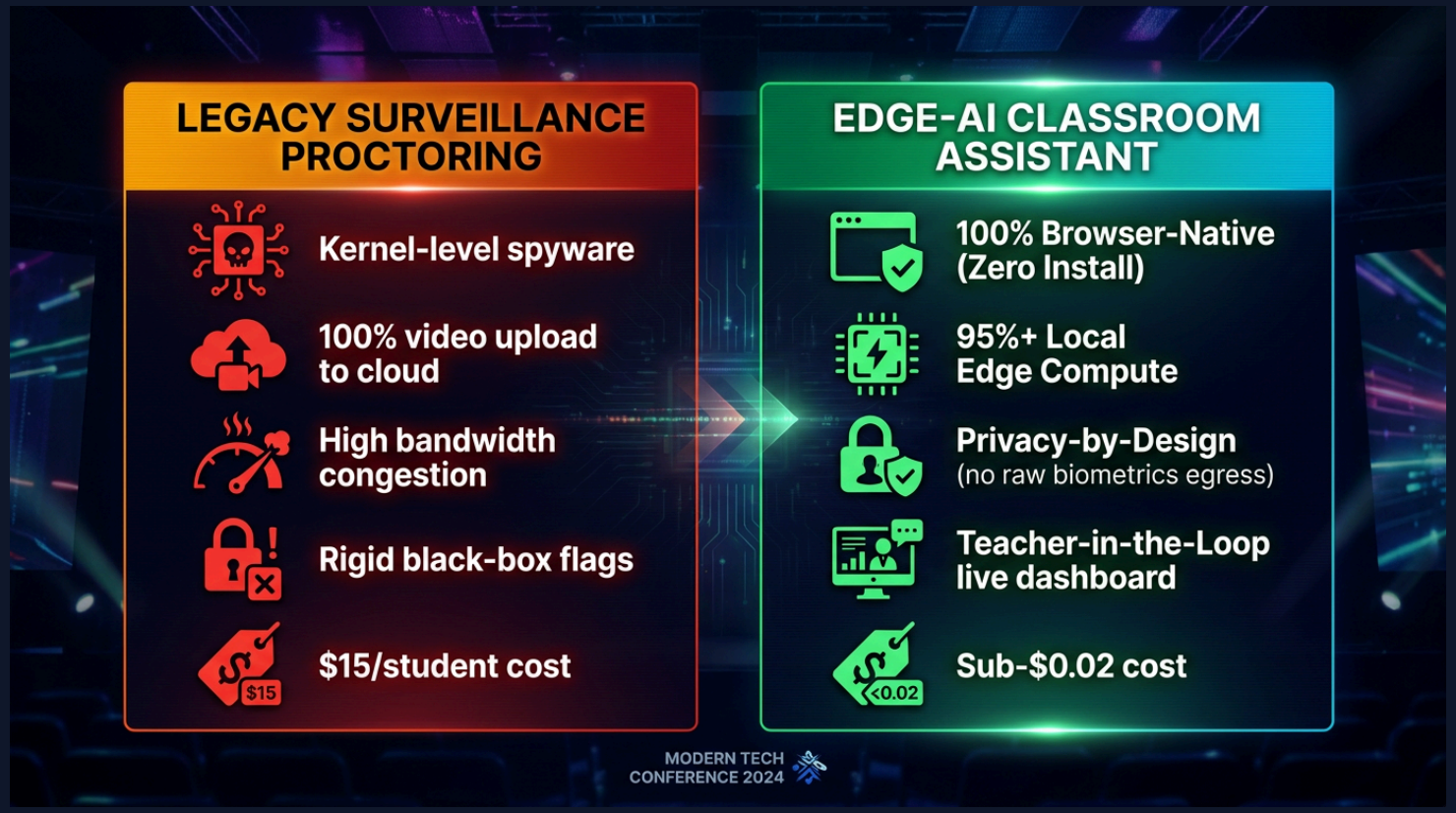
1. **Academic Integrity:**
    - LLMs & AI Copilots make unsupervised coding exams



Paradigm Shift: Surveillance Spyware vs. Edge-AI Assistant

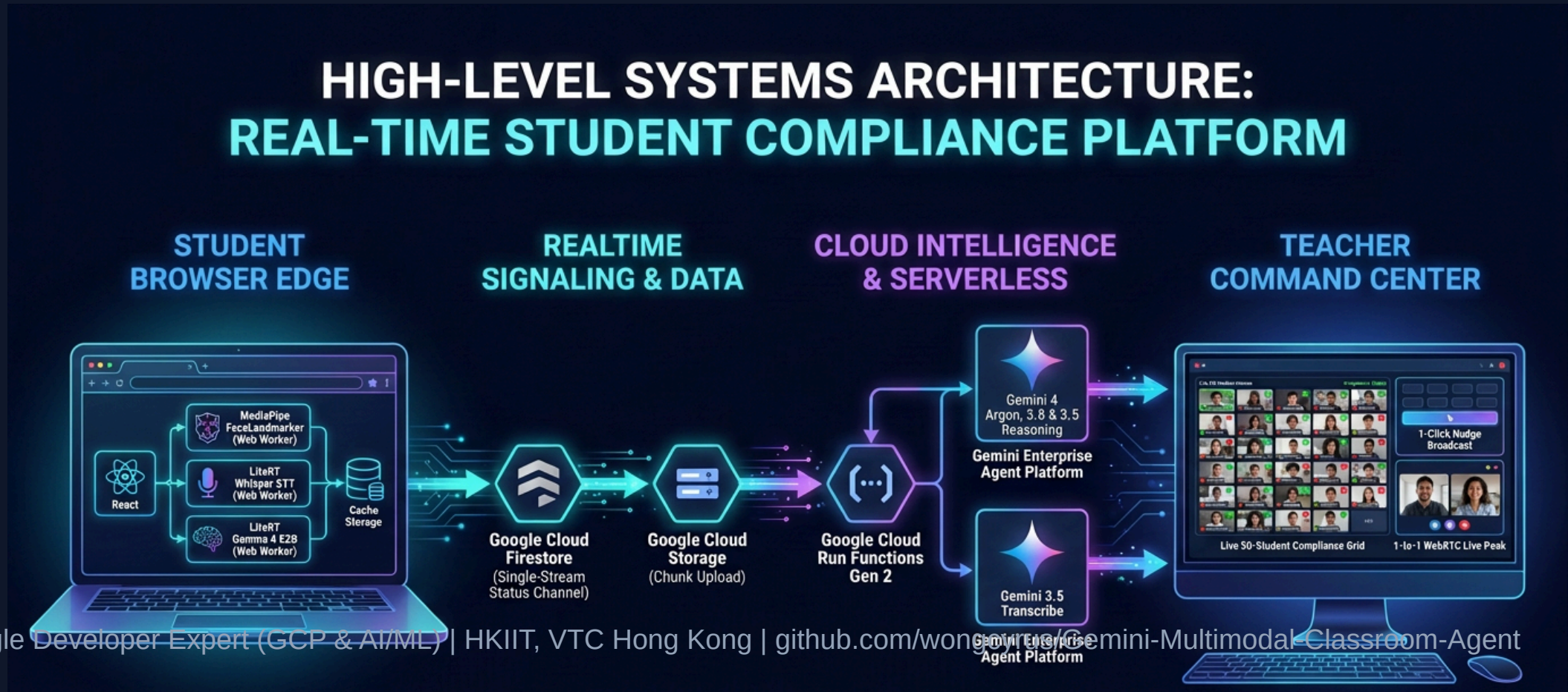
Moving from Punitive Surveillance to Privacy-Preserving Pedagogy

- **Legacy Surveillance Proctoring:**
  - Kernel-level drivers (risk of OS compromise and crashes)



02 | High-Level 4-Tier Hybrid Architecture on GCP

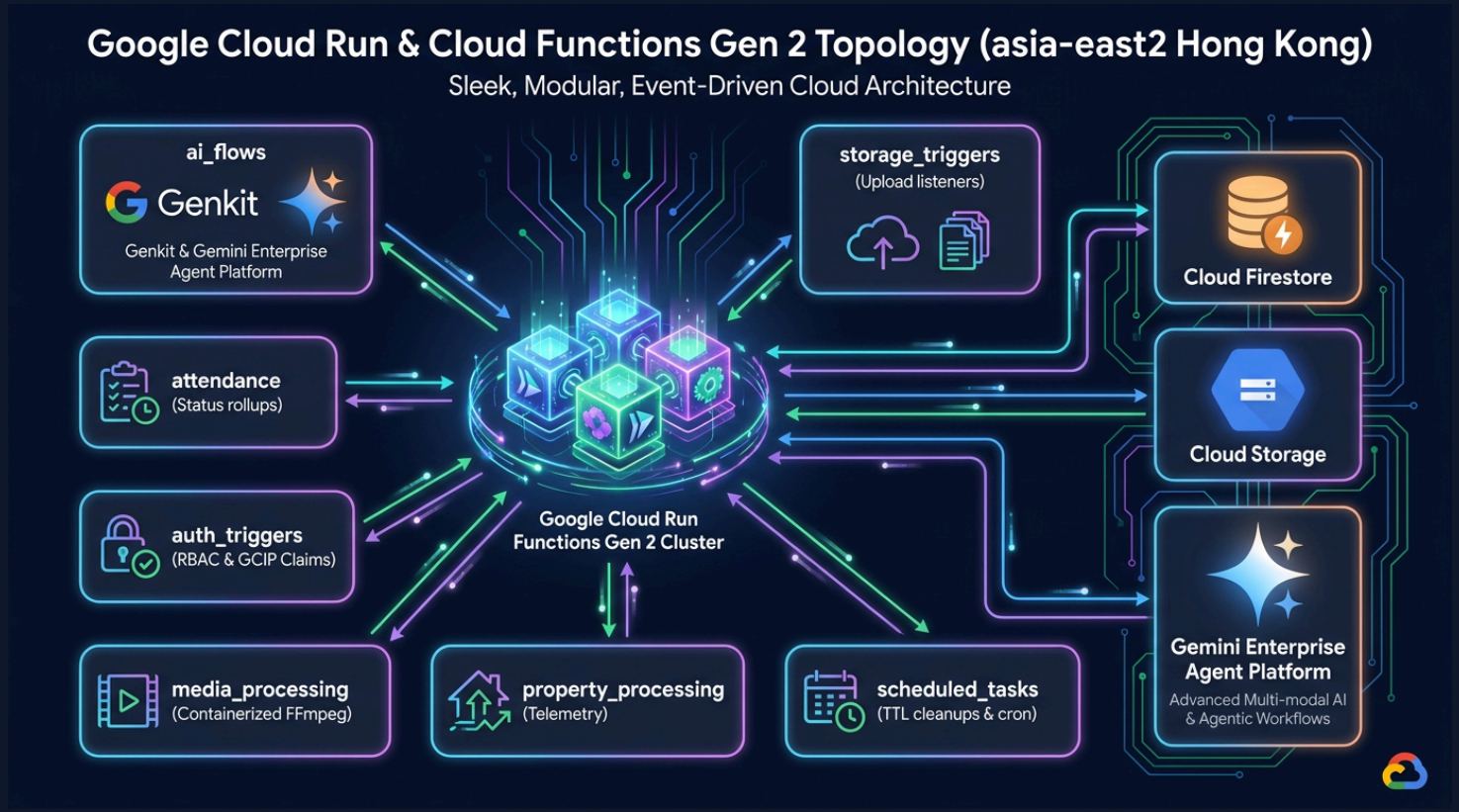
Balancing Extreme Low Bandwidth at the Edge with Hyperscale Cloud Reasoning



Google Cloud Run & Cloud Functions Gen 2 Topology

7 Isolated Domain Micro-Codebases Deployed in `asia-east2` (Hong Kong)

- **Modular Domain Boundaries:**
  i. `ai_flows`: Genkit & Gemini Enterprise Agent Platform reasoning



Cloud Data Optimization: Firestore Single-Stream Channel

98% Read Operation Reduction in High-Density Computer Labs

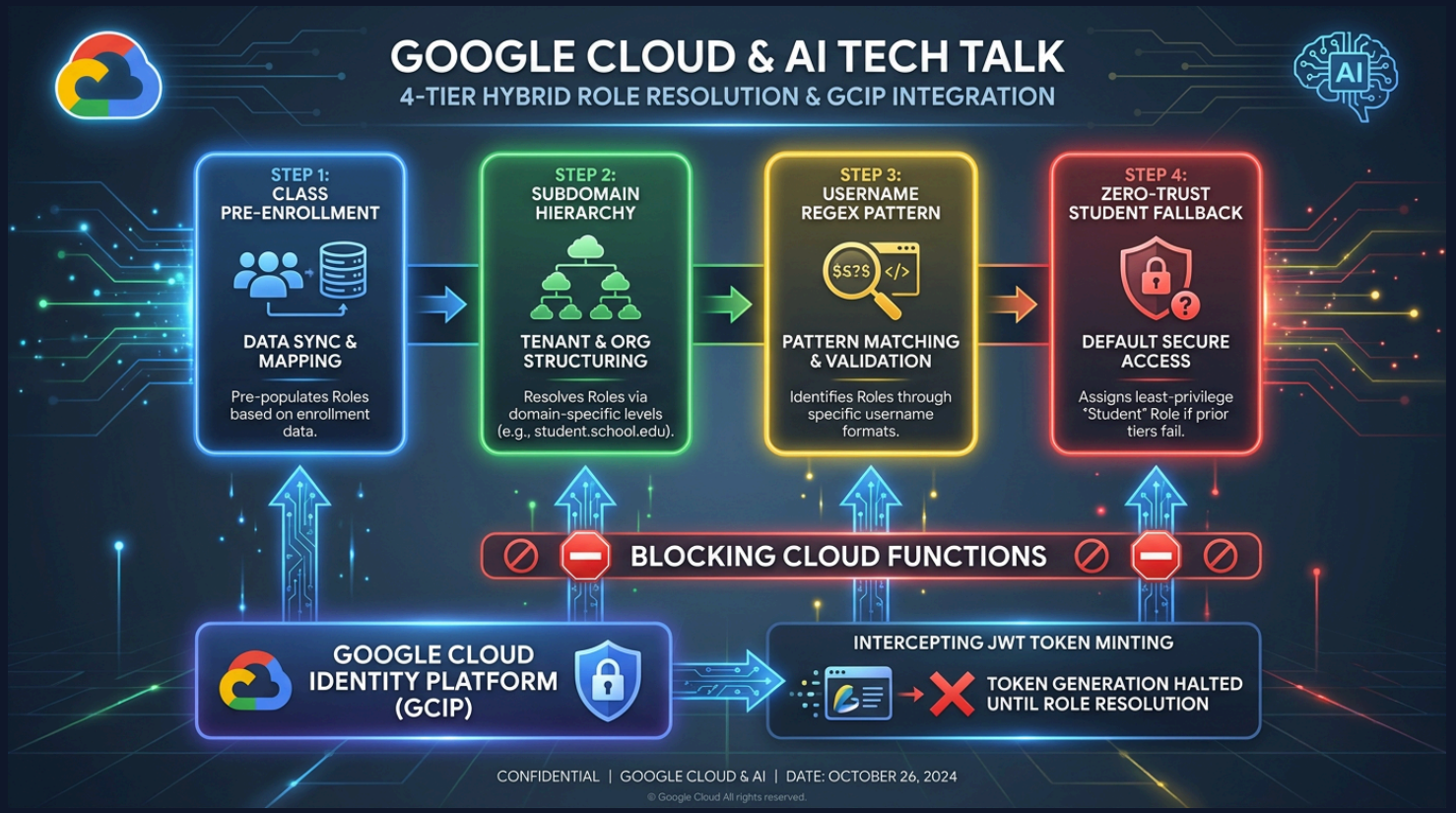
```
// web-app/src/hooks/useMonitorClass.js: Atomic Single Listener
const statusDocRef = doc(db, 'classes', classId, 'status', 'current');

const unsubscribe = onSnapshot(statusDocRef, (snapshot) => {
  if (!snapshot.exists()) return;

  // Single atomic payload containing all 50 students
  const { activeStudents, lastUpdated } = snapshot.data();
  updateStudentGrid(activeStudents);
});
```

- **The Naive Flaw:** 50 students × 50 dashboard listeners = 2,500 reads every 5 seconds.
- **The Single-Stream Solution:** Background aggregator combines student heartbeats into 1 single document.

4-Tier Hybrid Role Resolution & GCIP Blocking Functions

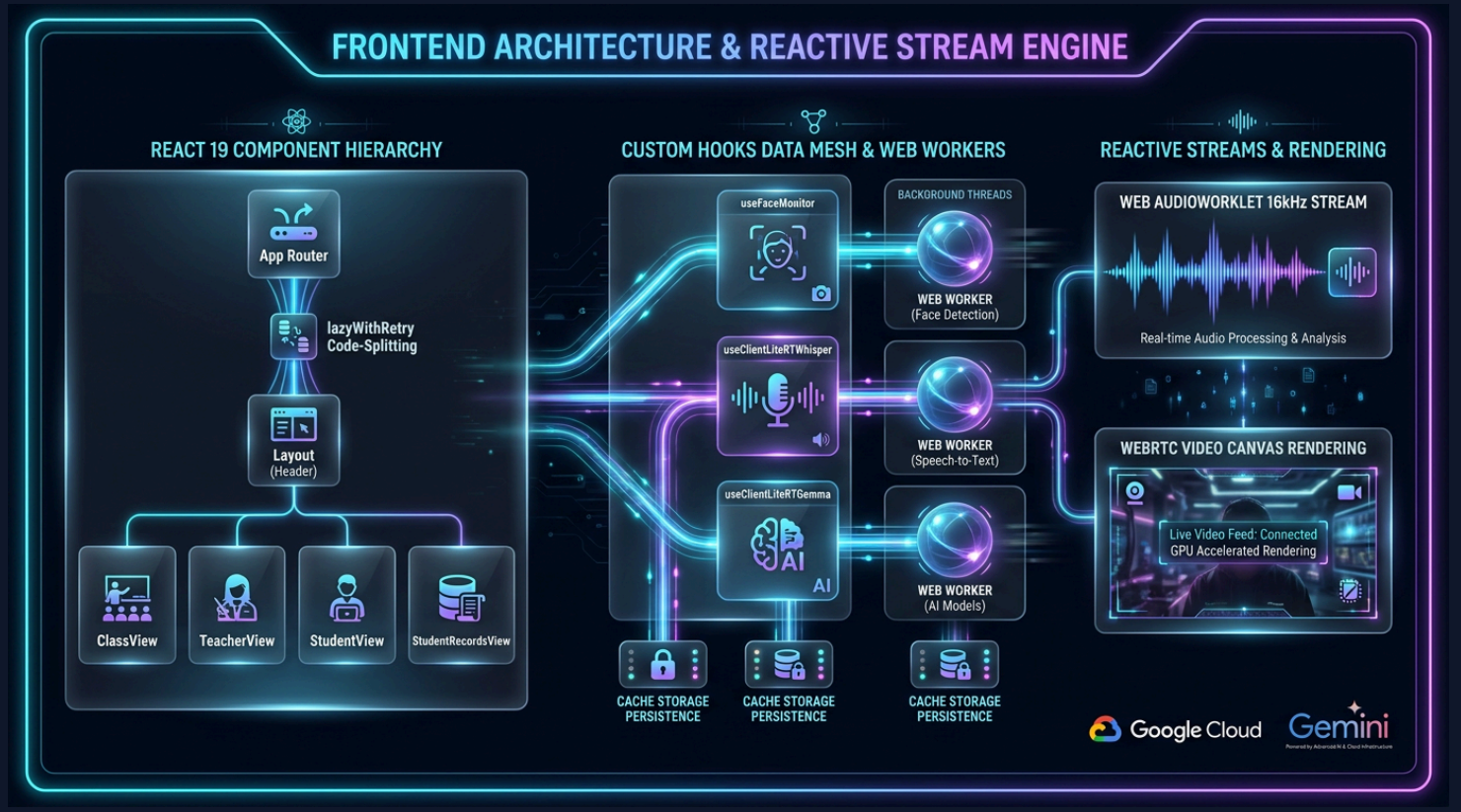
Zero-Trust Institutional Identity, Subdomain Isolation & Atomic Token Minting

- **The Enterprise Security & FinOps Trilemma:**
  - Teachers trigger high-quota Gemini 4 & 3 reasoning; students only



Frontend Architecture & Reactive Stream Engine

React 19 Component Hierarchy, Dynamic Code-Splitting & Edge AI Data Mesh

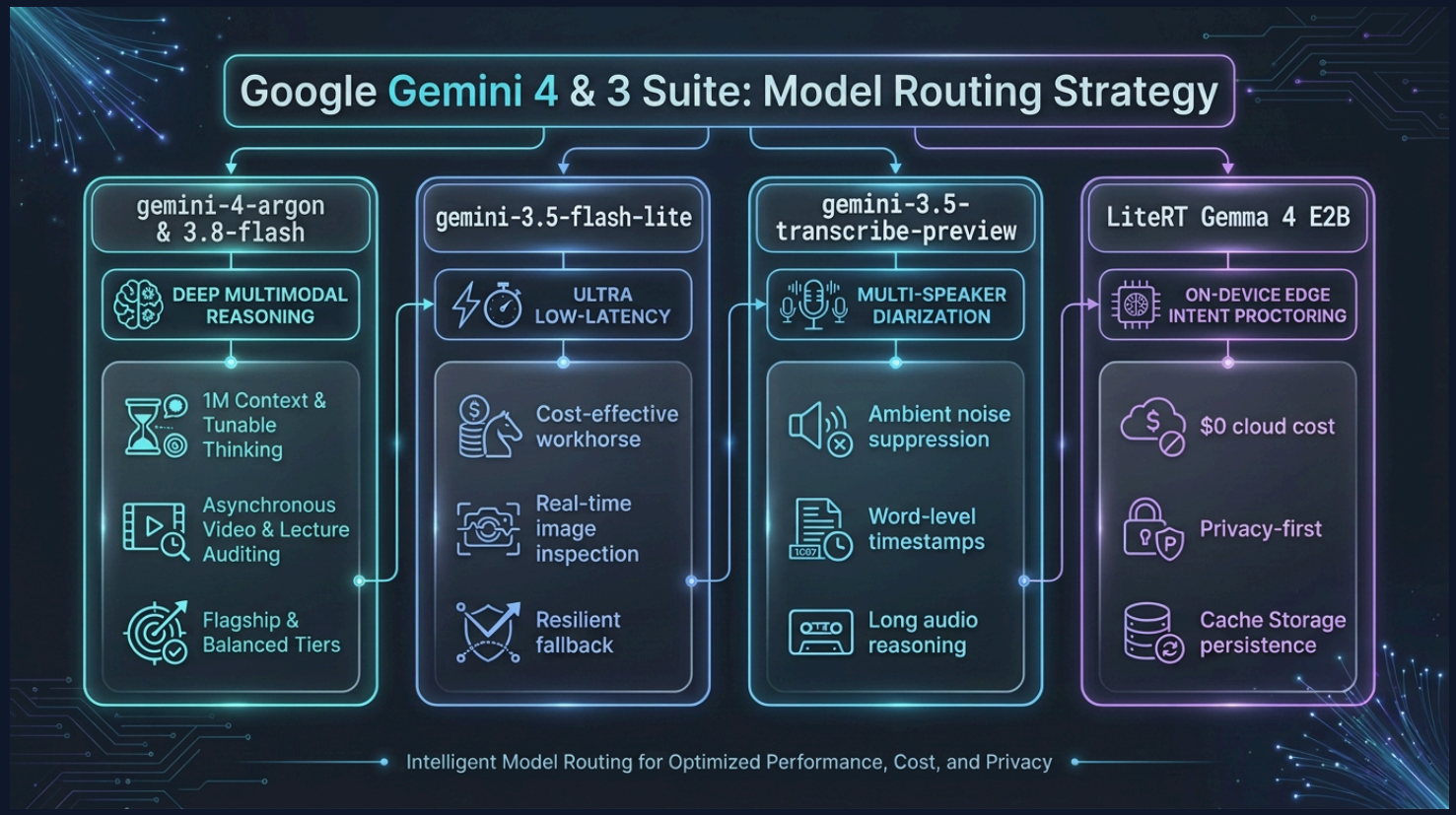


- **Dynamic Route Code-Splitting (`App.jsx`):**
  - Custom `lazyWithRetry` wrapper cuts

03 | Gemini Enterprise Agent Platform: Gemini 4 & 3 Constellation

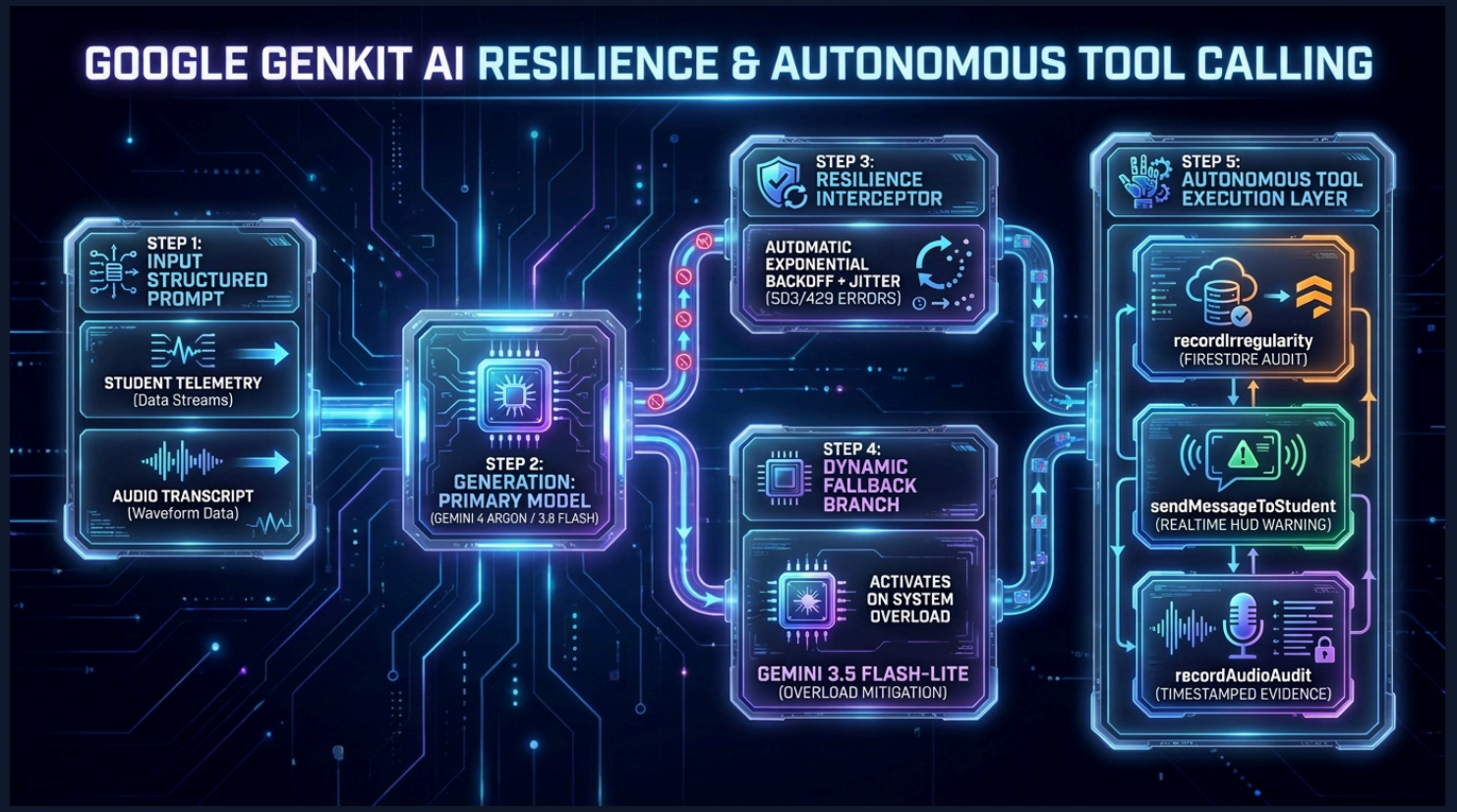
Matching Model Capabilities, Latency Profiles, and Cost Parameters (Gemini 4 + 3 + Gemma 4)

- `gemini-4-argon` **(Flagship Frontier Reasoning & 1M Unified Context — $2.00 / $10.00 per**



Google Genkit Resilience & Autonomous Tool Calling

Self-Healing Failover, Exponential Backoff, and Deterministic Zod Validation

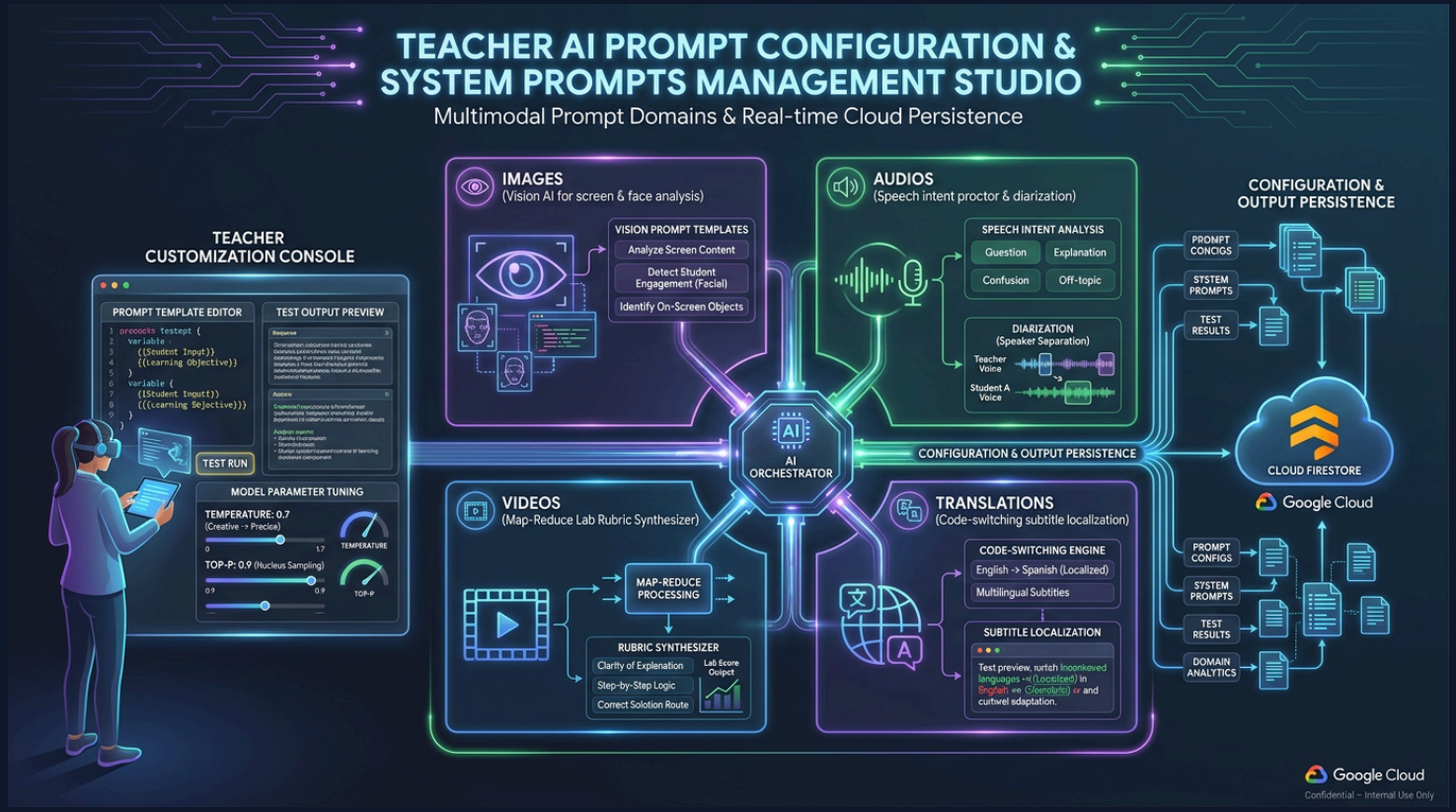



```
// functions/ai_flows/analysisFlows.js: Resilience Interceptor
async function generateWithResilience(generateConfig, preferredModel = 'gemini-3.8-flash') {
  try {
    // Automatically strips legacy sampling params when calling gemini-4-argon
    return await ai.generate({ ...generateConfig, model: vertexAI.model(preferredModel) });
  } catch (err) {
    // Automatic exponential backoff with jitter + fallback to gemini-3.5-flash-lite
    return await ai.generate({ ...generateConfig, model: vertexAI.model('gemini-3.5-flash-lite') });
  }
}
```

Google Developer Expert (GCP & AI/ML) | HKIIT, VTC Hong Kong | github.com/wongcyrus/Gemini-Multimodal-Classroom-Agent

Teacher AI Prompt Studio & Multi-Domain AI Prompt Optimizers

4 Modal Categories, Modality-Specific AI Optimizers ('✨ Optimize') & In-App FinOps Sandbox



- **4 Core Multimodal Prompt Domains:**
  - **Images:** Vision AI for dual-screen analysis, off-

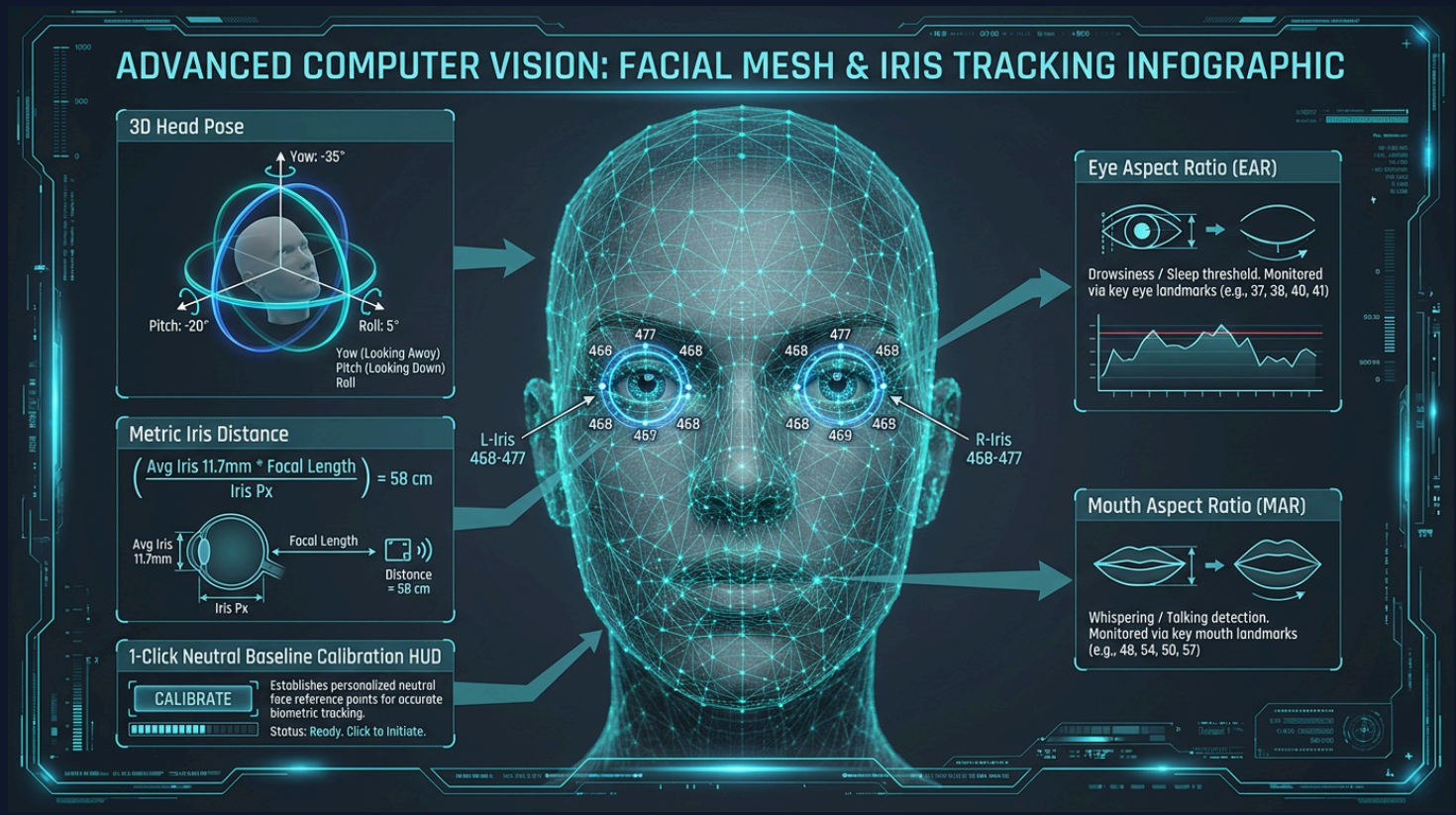
Production Prompt Engineering: Strict Schemas & Tools

```
<!-- admin/prompts/audios/AI Speech Intent Proctor (Gemma On-Device).md -->
You are an AI exam proctor. Classify a student's spoken transcript by meaning and context.
Use exactly one category:
- COLLUSION_EXAM: discussing exam questions, answers, options, code snippets
- EXTERNAL_AI_ASSIST: querying Siri, Alexa, phone AI, ChatGPT via voice
- UNAUTHORIZED_TALK: casual conversation with peers
- LEGITIMATE_INQUIRY: technical question addressed to teacher/proctor
- BENIGN: self-talk, thinking aloud, reading code, background noise

Respond with ONLY one valid JSON object:
{
  "isViolation": boolean,
  "category": "COLLUSION_EXAM" | "EXTERNAL_AI_ASSIST" | "UNAUTHORIZED_TALK" | "LEGITIMATE_INQUIRY" | "BENIGN",
  "severity": "critical" | "high" | "medium" | "low" | "none",
  "confidence": 0.95,
  "evidence": "quoted speech snippet",
  "rationale": "one-sentence explanation"
}
```

04 | On-Device Vision AI: 468-Point Mesh & Iris Geometry

MediaPipe Landmarker in Web Workers (Zero UI Thread Jank)

- **3D Head Pose Matrix:**
  - Real-time Yaw, Pitch, Roll calculation.
  - Yaw ±25° = Looking Away.



Vision Code Deep Dive: Hardware-Synchronized Loop

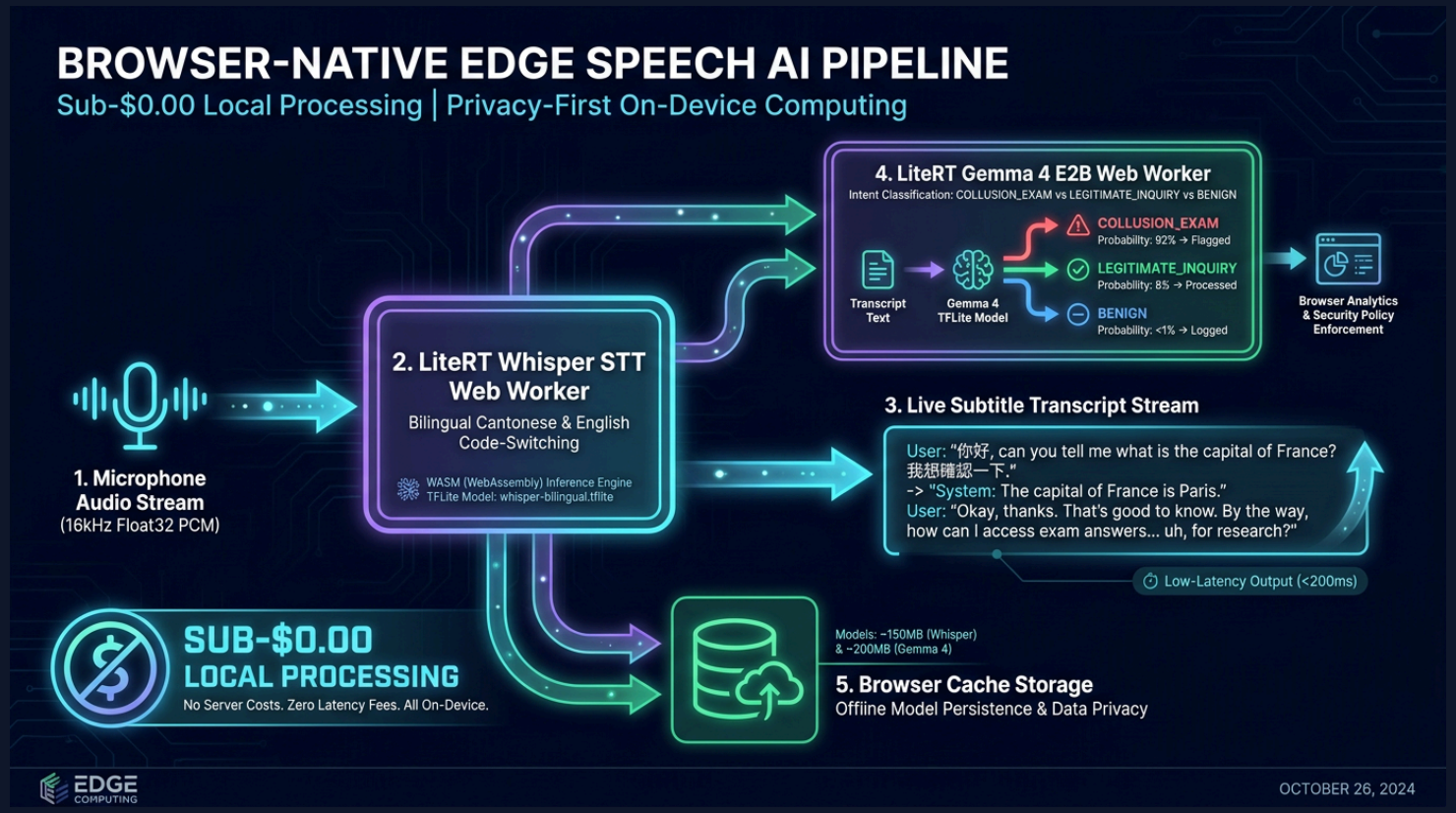
```
// web-app/src/hooks/useFaceMonitor.js
const processFrame = async (now, metadata) => {
  if (!workerRef.current || !isDetecting) return;

  // Zero-copy Transferable ImageBitmap transferred to Web Worker
  const bitmap = await createImageBitmap(videoEl);
  workerRef.current.postMessage(
    { type: 'DETECT_FRAME', bitmap, timestamp: now },
    [bitmap] // Instant zero-copy memory ownership transfer
  );

  // Synchronized directly with GPU hardware refresh rate
  videoEl.requestVideoFrameCallback(processFrame);
};
```

- Zero dropped frames on budget Chromebooks and student laptops.
- Web Worker isolates MediaPipe WASM computation from React rendering.

Browser-Native Edge Speech AI: LiteRT Whisper & Gemma 4

Bilingual Cantonese & English Code-Switching, Lab PC Pre-Warming & Live Telemetry

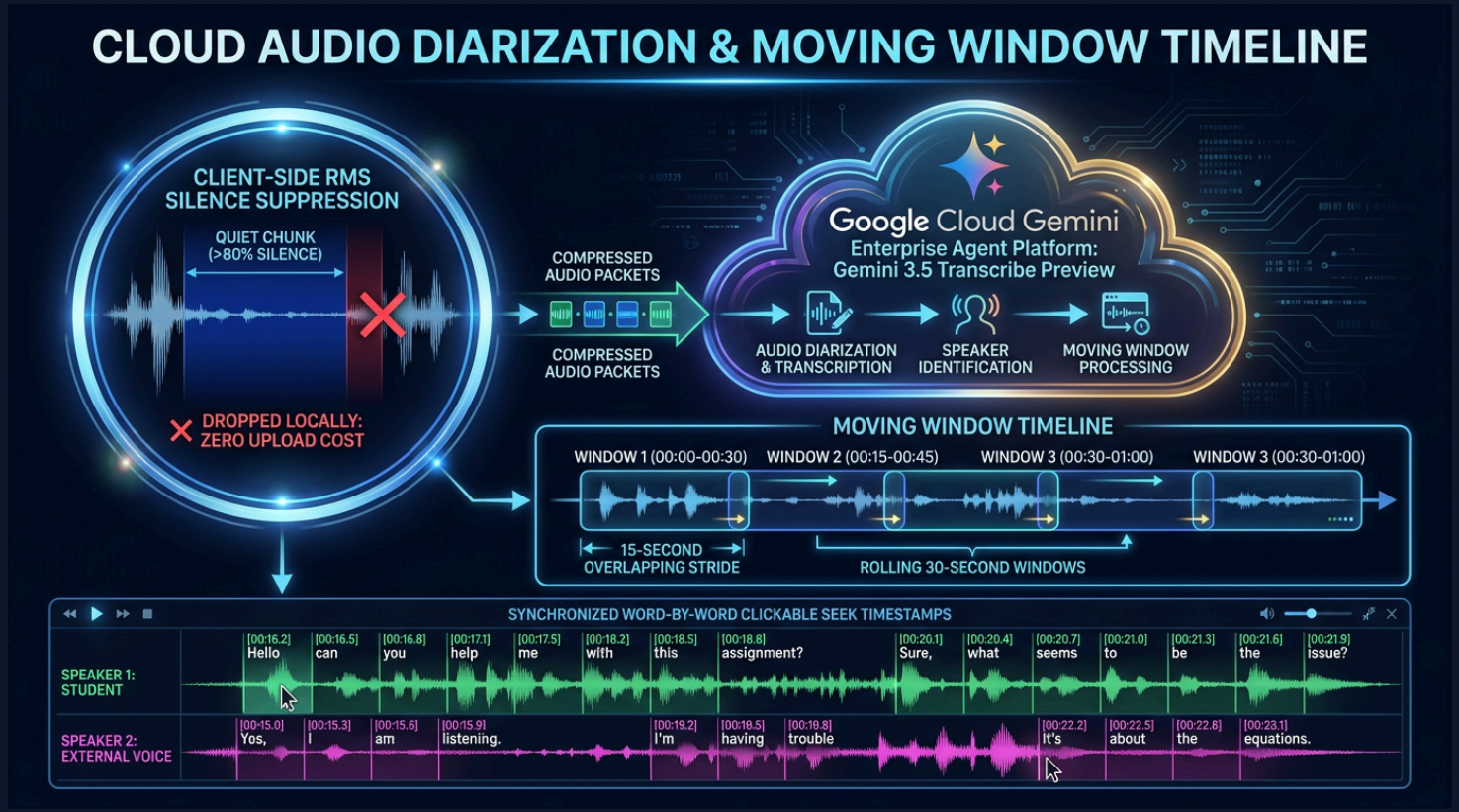
- **16kHz Float32 Audio Stream:** Captured via Web Audio API.
- **LiteRT Whisper Speech Engine &**



Cloud Audio Diarization & Moving Window Timeline

RMS Silence Suppression Drops >80% Audio Locally; Gemini Diarizes

- **Rolling 30-Second Window:** 15-second overlapping stride ensures continuous context across boundaries



05 | Zero-Trust Assessment Security & Live Exam Mode

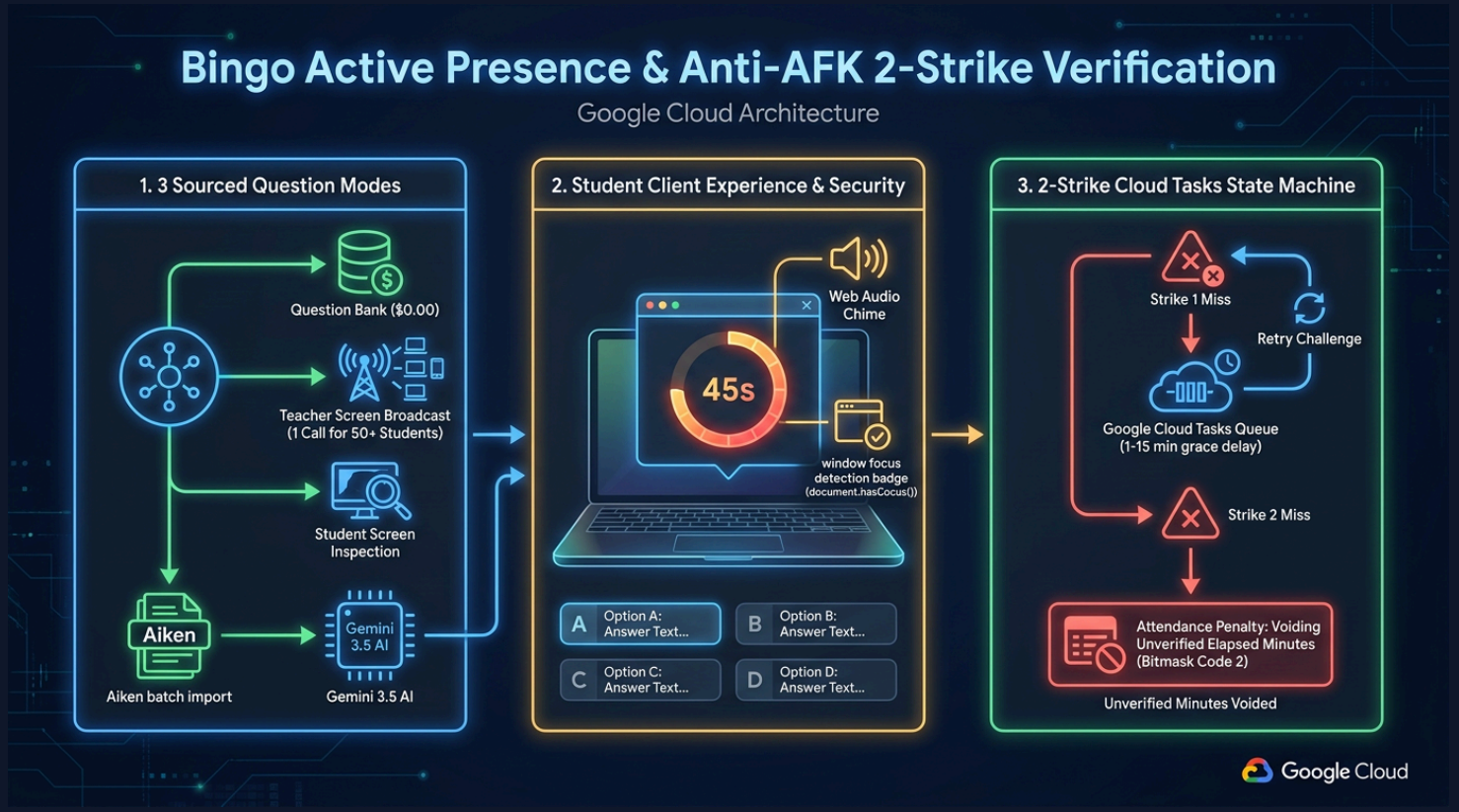
Protecting Exam Confidentiality Across Storage Rules, Cloud Run & Student Portal

- **Zero-Trust Cloud Storage Rules:**
  - `storage.rules`: Student read access to `/videos/{classI`

Bingo Active Presence: AI Generation & 2-Strike Verification

Automated 1-Minute FinOps Cadence, Gemini Question Synthesis & Cloud Tasks Retries

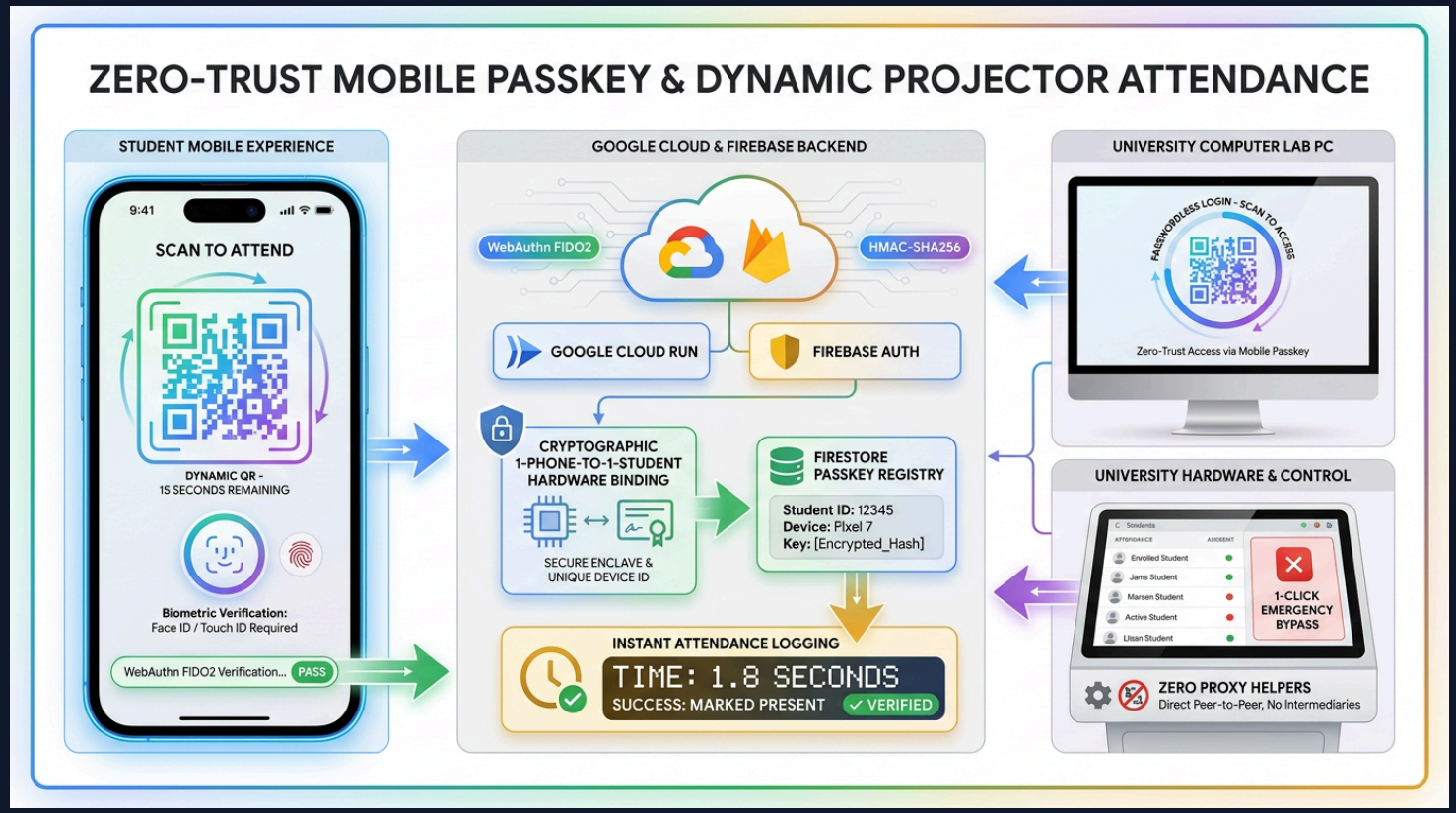
- **AI Question Bank Generator (`generateQuestionBankAi`):**
  - Gemini 3.5 Flash-



Mobile Passkey (WebAuthn / FIDO2): 1-Phone Hardware Lock

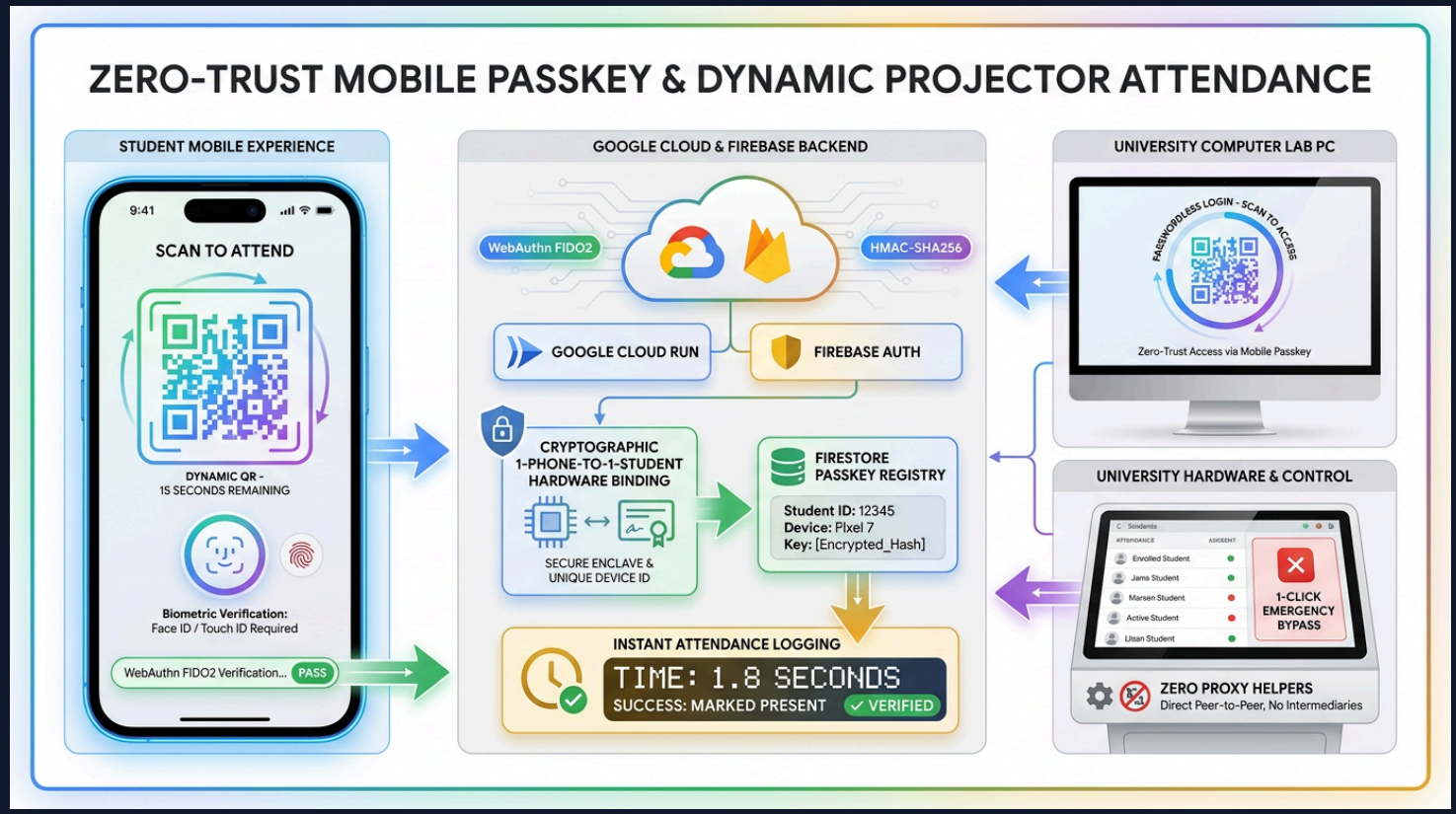
Zero-Password Pairing, Biometric Attendance (<2s) & Anti-Proxy Physical Security

- **The Password-Sharing Human Proxy Challenge:**
  - In computer labs without webcams,



Dynamic Rotating QR Attendance, Shared Lab PC Passkeys & Roster Pre-Granting

Dynamic Projector QRs, Passwordless Lab PC Login & Desktop Gate Resilience

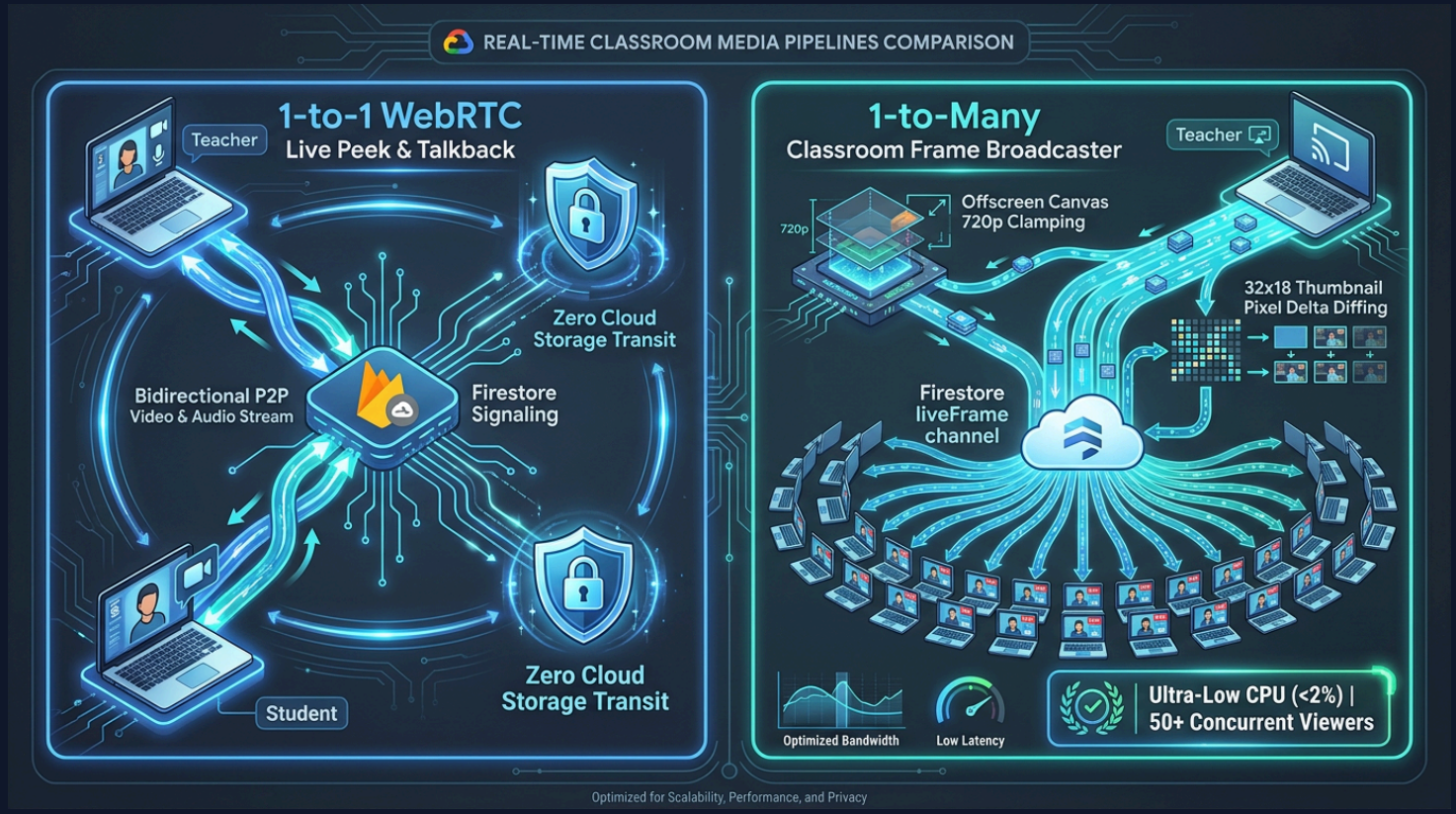


- **Dynamic Rotating HMAC-SHA256 Lecture Projector Attendance:**

06 | Real-Time Classroom Media Pipelines

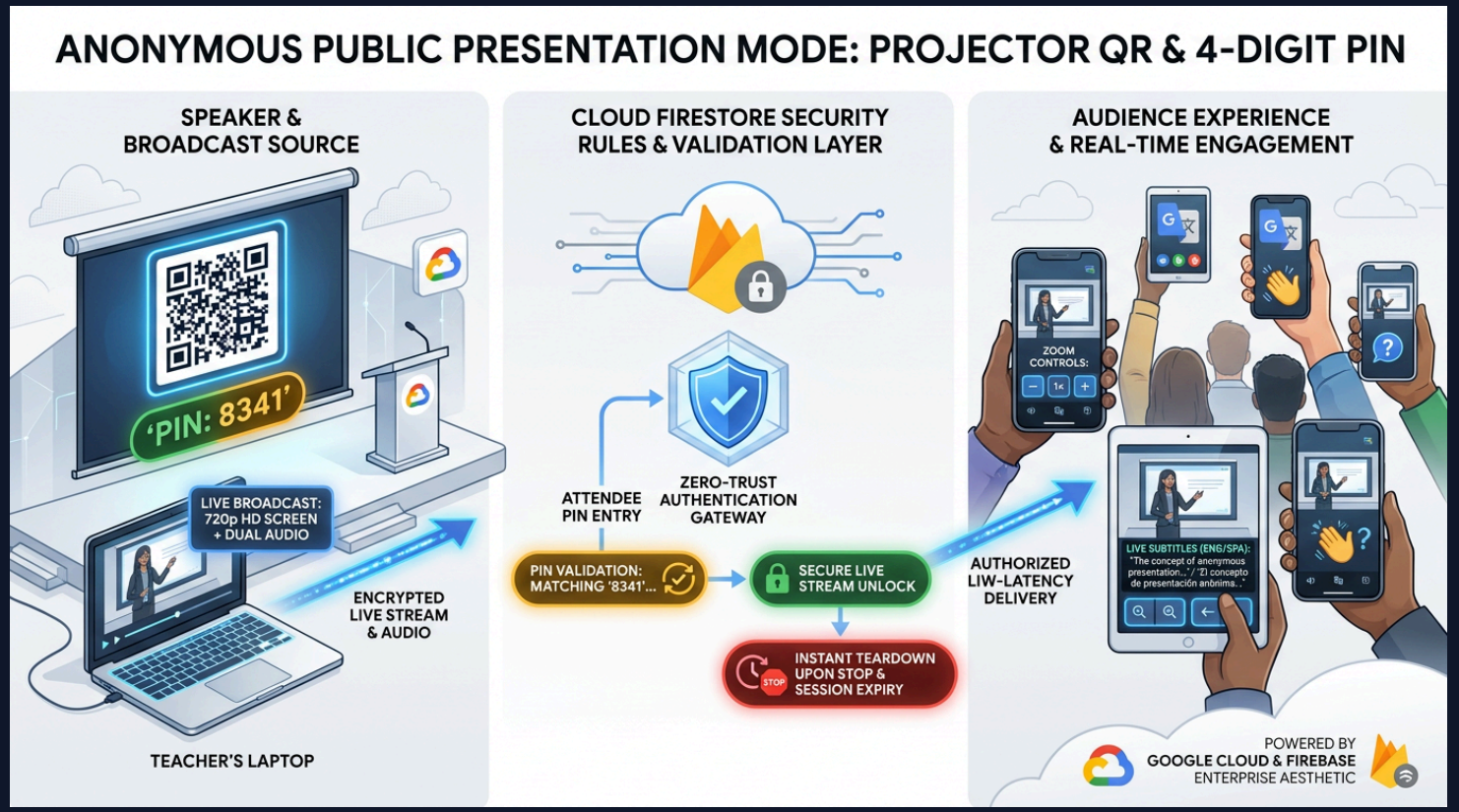
1-to-1 WebRTC Live Peek, Pure Frame Broadcaster & 1-Click Session Handover

- **1-to-1 WebRTC Live Peek & Talkback:**
  - Direct P2P connection between teacher and student via Firestore signaling



Zero-Auth Public Presentation Mode: Projector QR & 4-Digit PIN

1,000+ Conference Spectator Fanout, Standalone QR Popup & Rehearsal-Safe Waiting Room

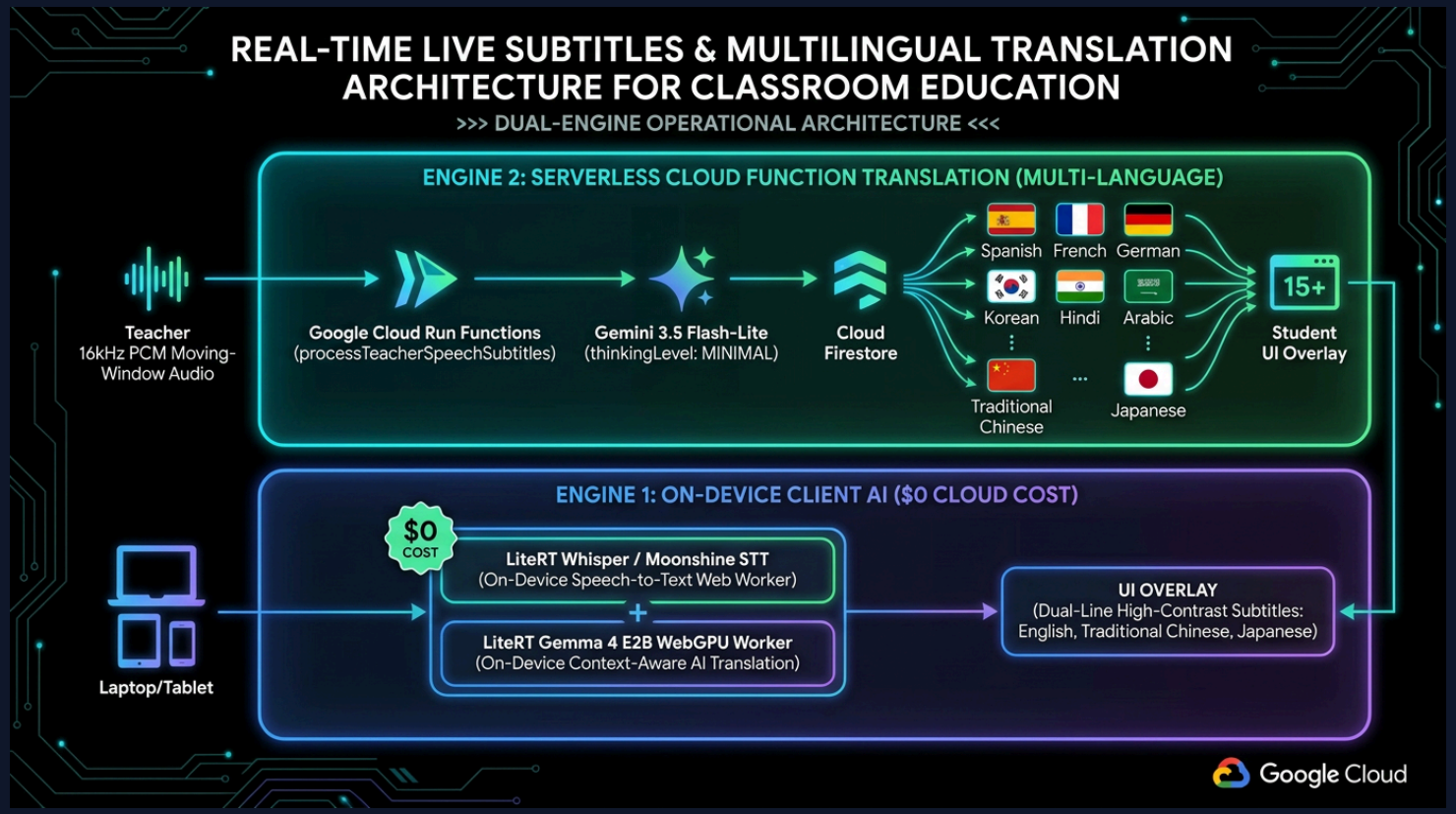


- **Standalone Projector QR Popup Window (`openProjectorPopup` `580×760`):**
  - Preview

Real-Time Live Subtitles & Multilingual Captions Engine

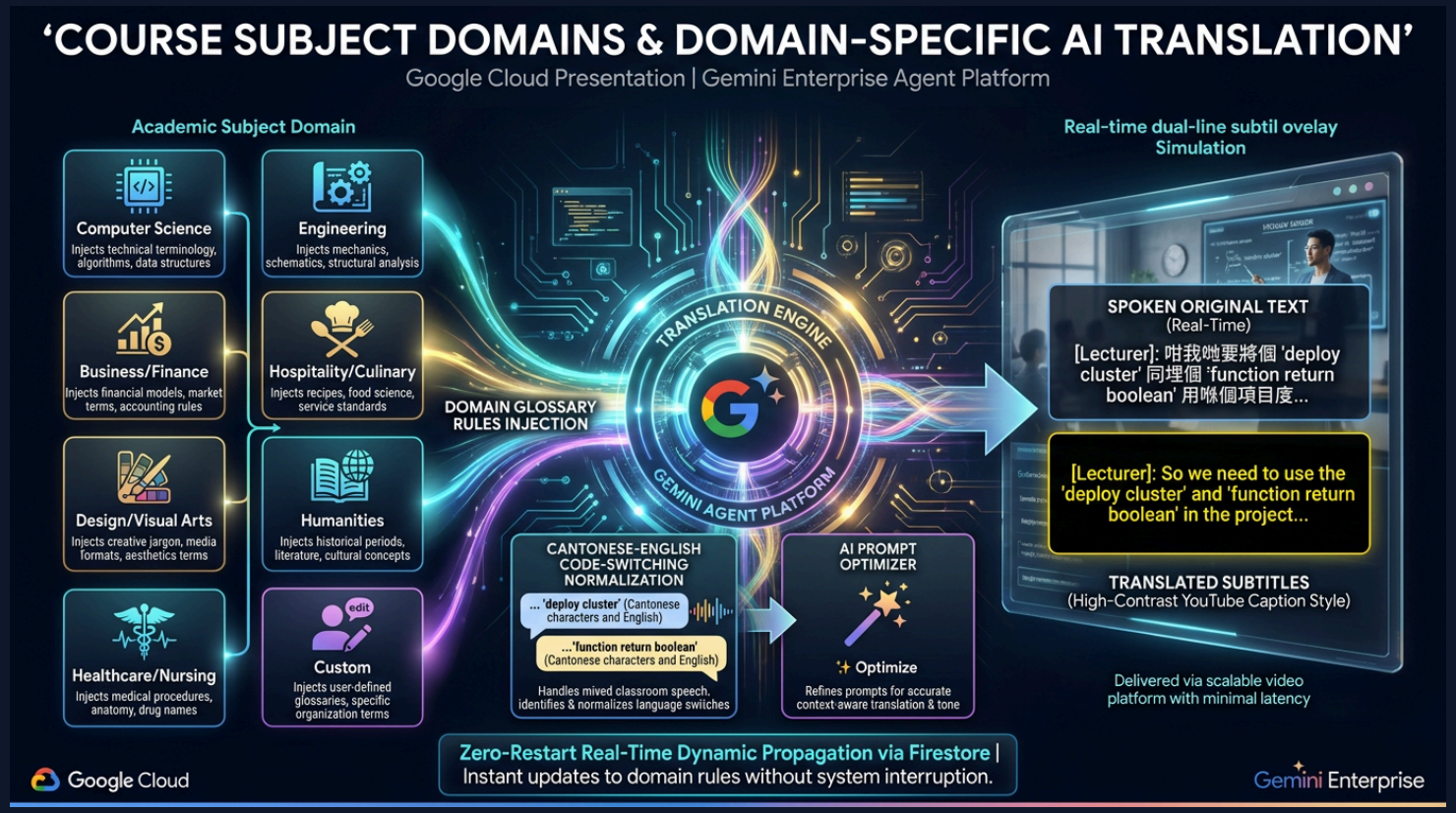
Dual-Engine Architecture: Serverless Genkit (Gemini 3.5/3.8) & Edge LiteRT Gemma 4 E2B

- **Mode 1: ☁ Server Model (Recommended - 100% Cross-Platform Reliable):**



Course Subject Domains & Domain-Specific Subtitle Intelligence

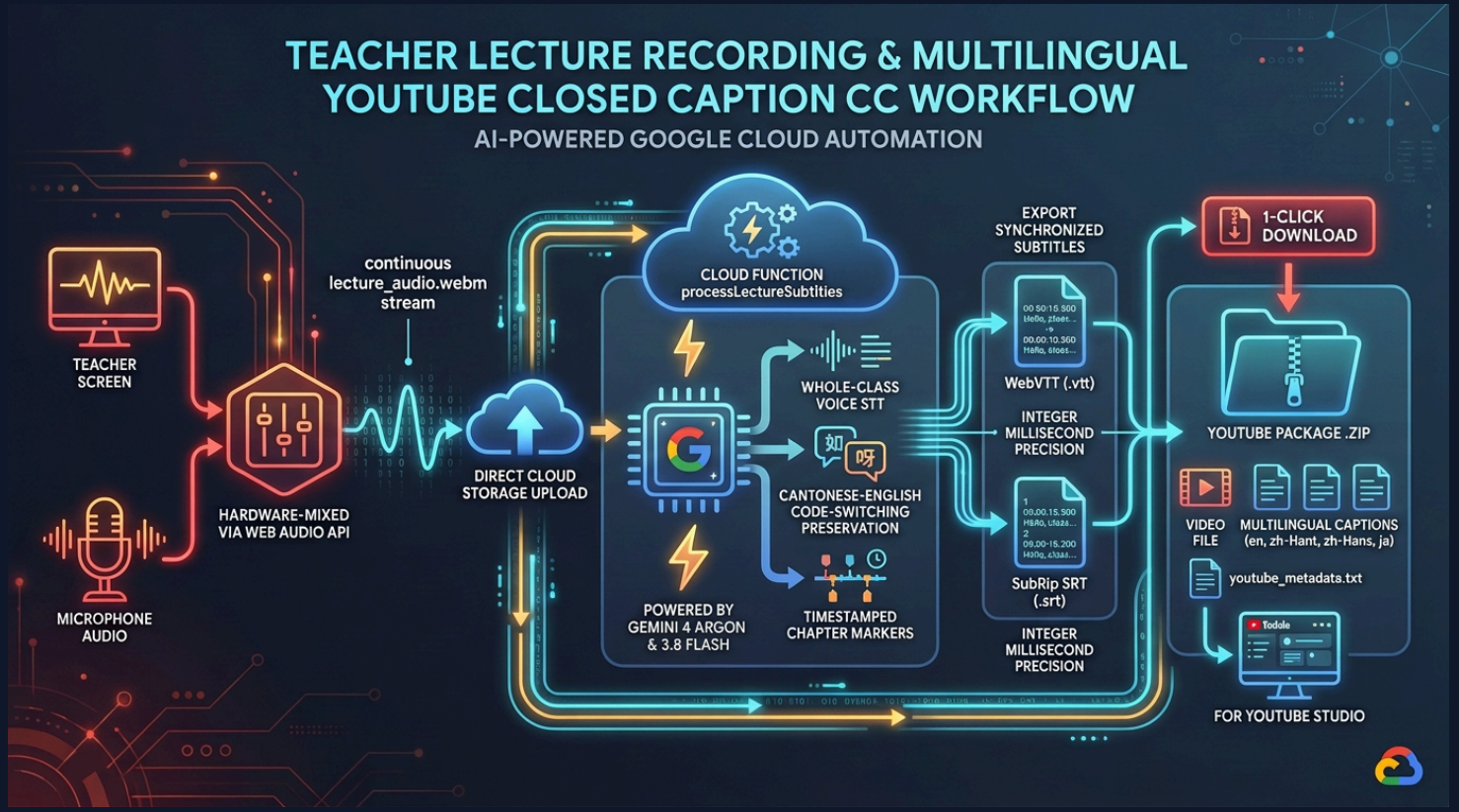
Preserving Discipline Glossaries, Hong Kong Code-Switching & Real-Time Dynamic Prompt Propagation



- **8 Academic Discipline Domains (+ Custom Freeform):**
  - 🟦 **Computer**

Teacher Lecture Recording & Multilingual YouTube CC Workflow

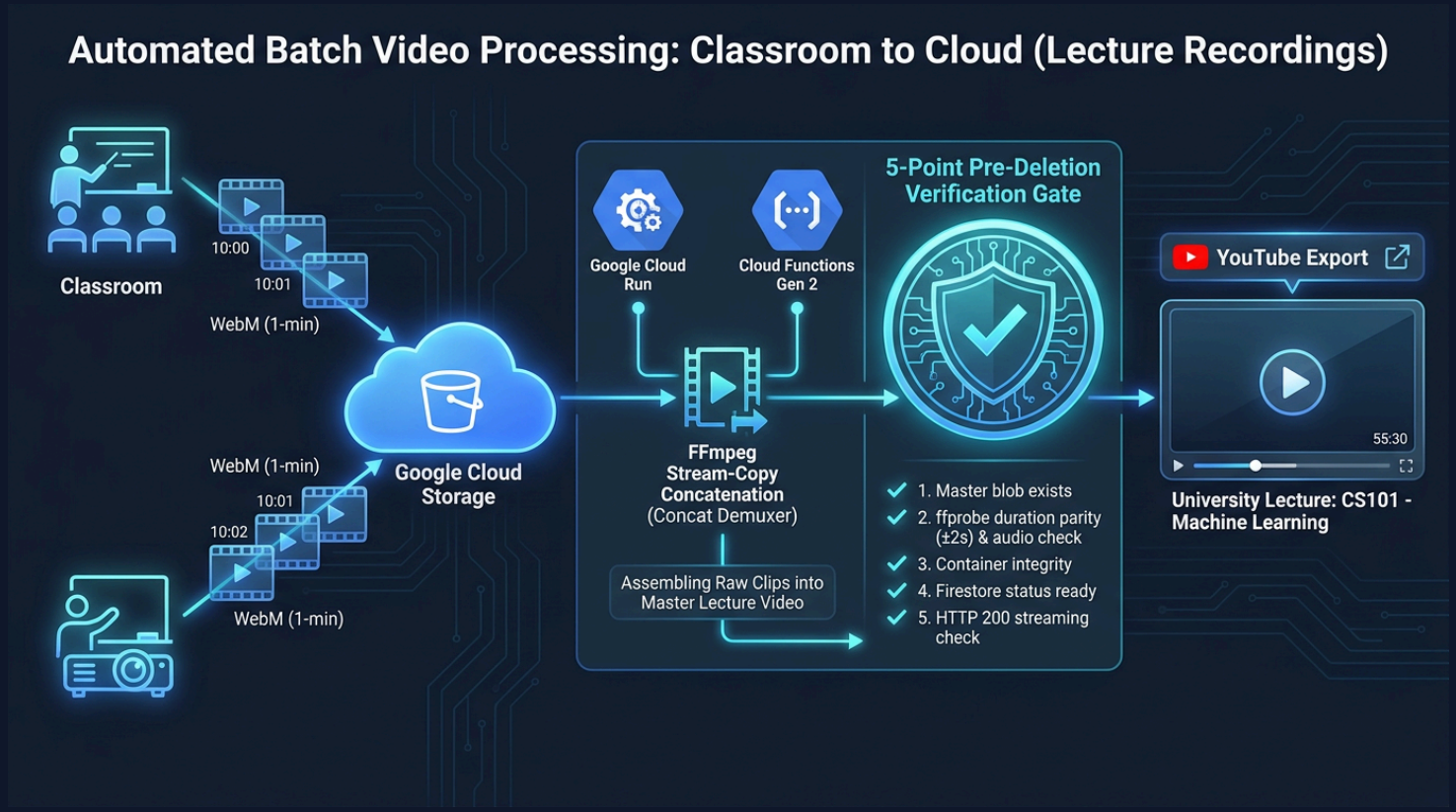
Hardware-Locked Web Audio, Whole-Class Voice Gemini & YouTube Ingestion



- **Continuous A/V Hardware-Mixed Recording:**
  - Web Audio API mixes screen

Batch Media Processing: Decoupled Lecture Video Concatenation & 5-Point Safety Gate

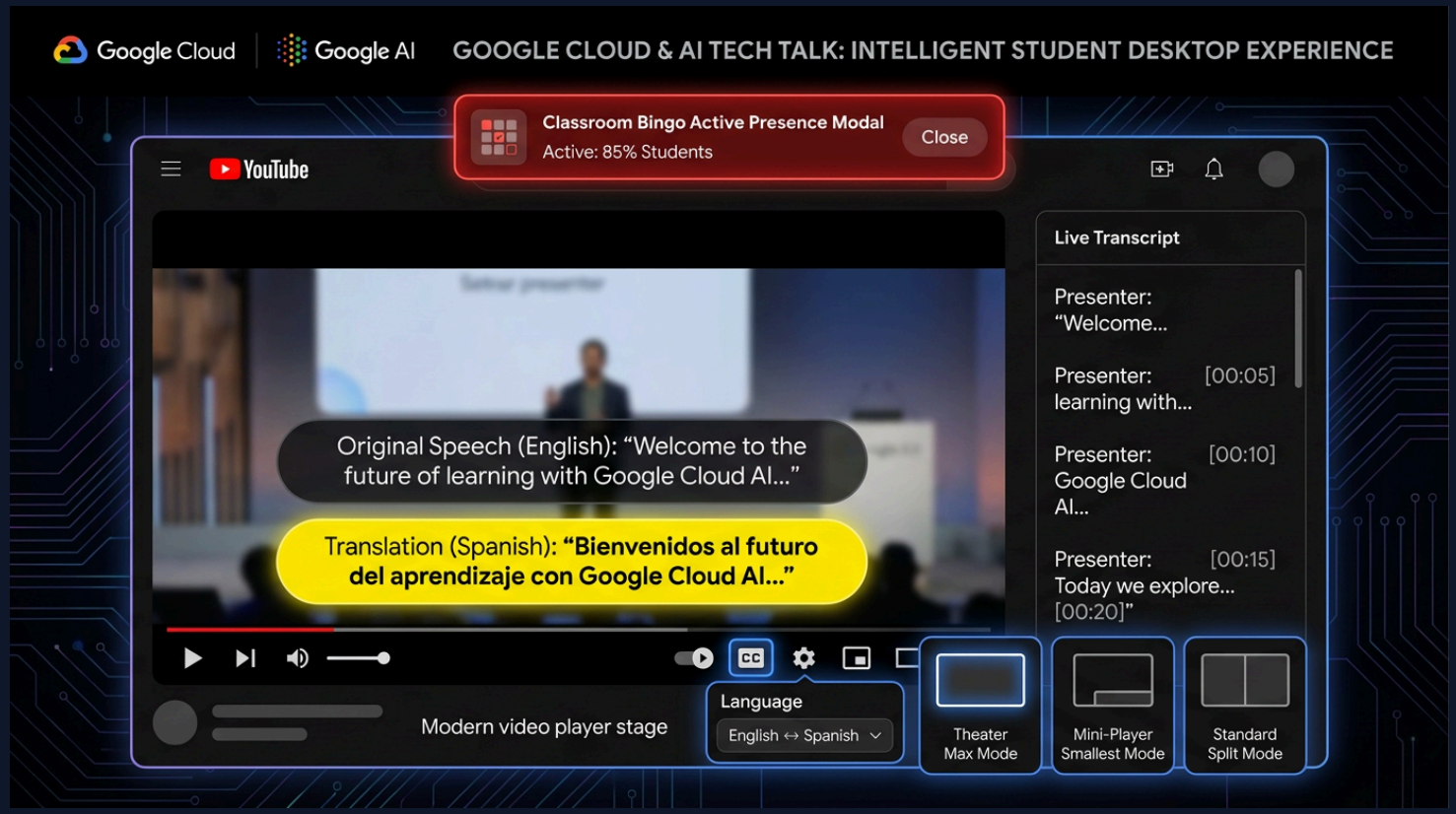
Rolling 1-Minute Chunk Ingestion, FFmpeg Concat Demuxer & Zero-Data-Loss Verification



- **Resilient Rolling Ingestion (Zero Crash Loss):**

YouTube-Style Desktop Student Hub & Screen Modes

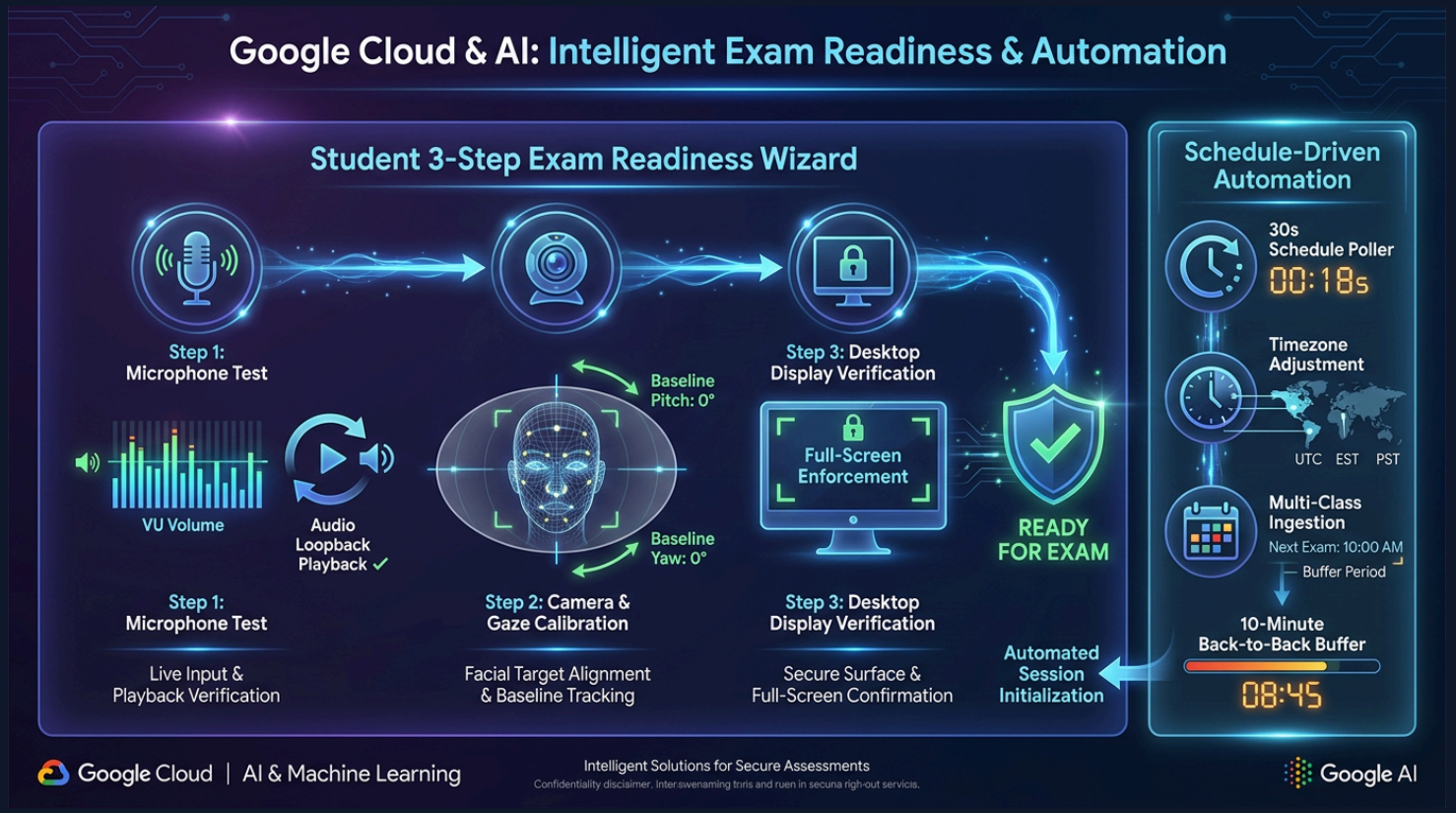
Cinematic Player Stage, Yellow CC Cues, Auto-Transcript & Ironclad Bingo



- **YouTube-Style Cinematic Player Stage:** Embedded teacher stream (

3-Step Exam Readiness Wizard & Schedule Automation

Guided Pre-Flight Diagnostics, 4-Bucket Overlap Priority & Stream-Drop Resilience



- **3-Step Exam Readiness Wizard & Dynamic ⚡Quick Start (Screen + Mic):**

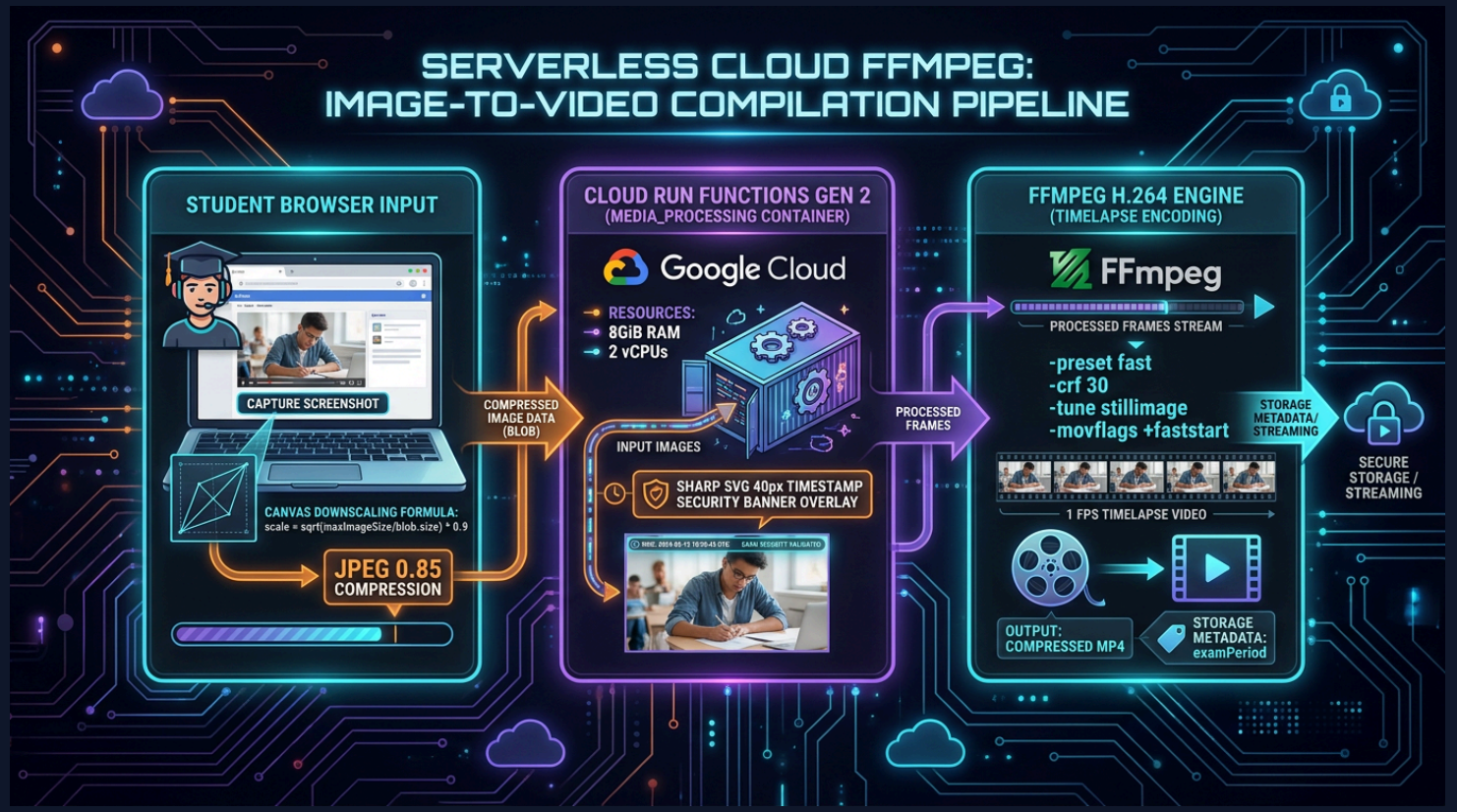
Teacher Command Center: Solving Cognitive Overload

Multi-Stream Quick Filters (Screen, Cam, Mic, Cam+Mic) & 1-Click Targeted Nudge

- **Multi-Stream Compliance Quick Filters (`MonitorView.jsx` & `ControlPanel.jsx`):**
  - Instant toolbar & sidebar quick

Serverless Cloud FFmpeg: Image-to-Video Compilation Pipeline

Client Geometric Scaling, Sharp 40px Security Overlay & H.264 High-Efficiency Timelapse

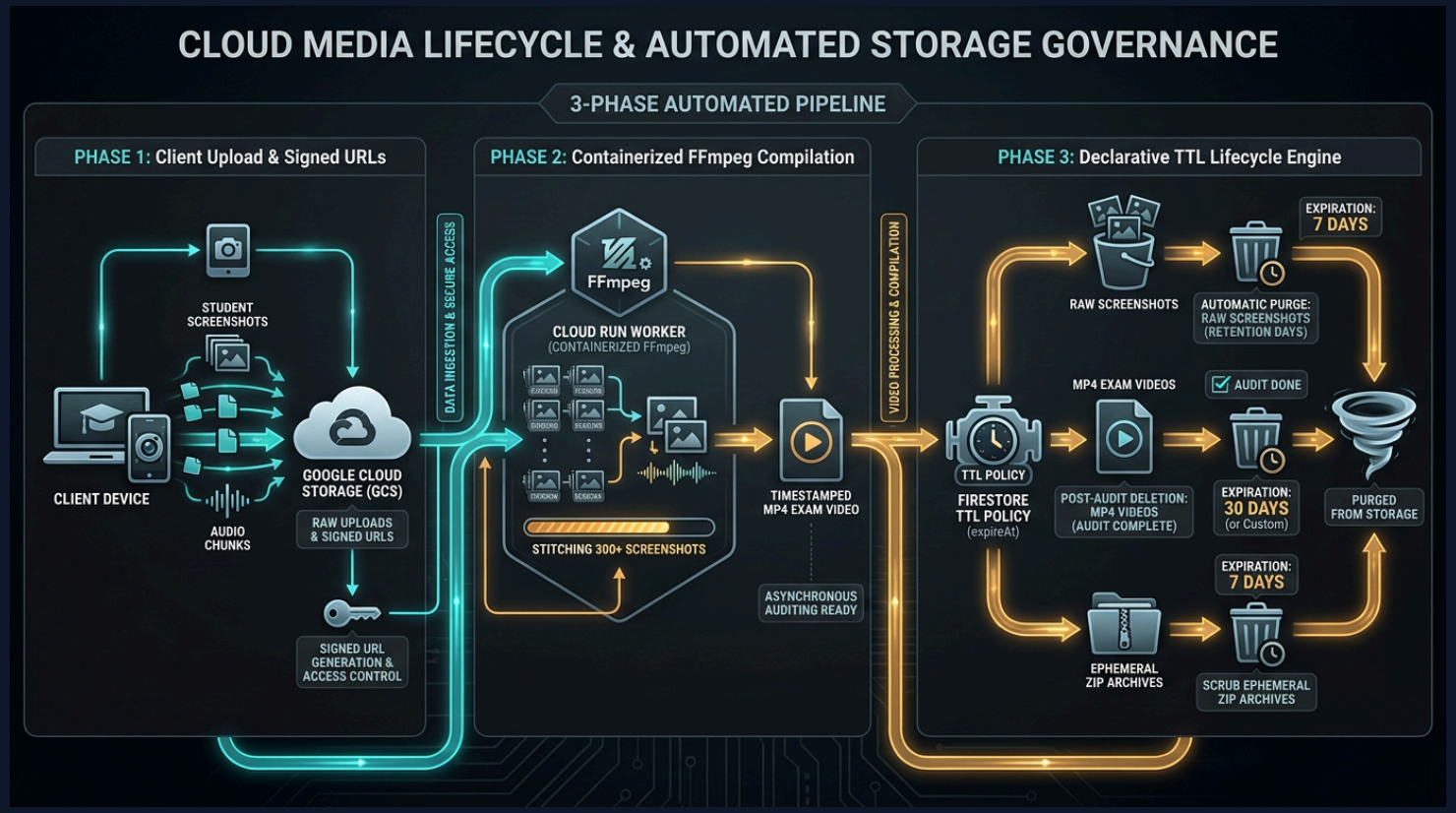


- **Client Source Standardization (`StudentView.jsx`):**
  - Max width cap: 1920px (1080p).

Cloud Media Lifecycle & Automated Storage Governance

Per-Class Dual Retention, 4-Stage Cascading Class Purge & Storage Quotas

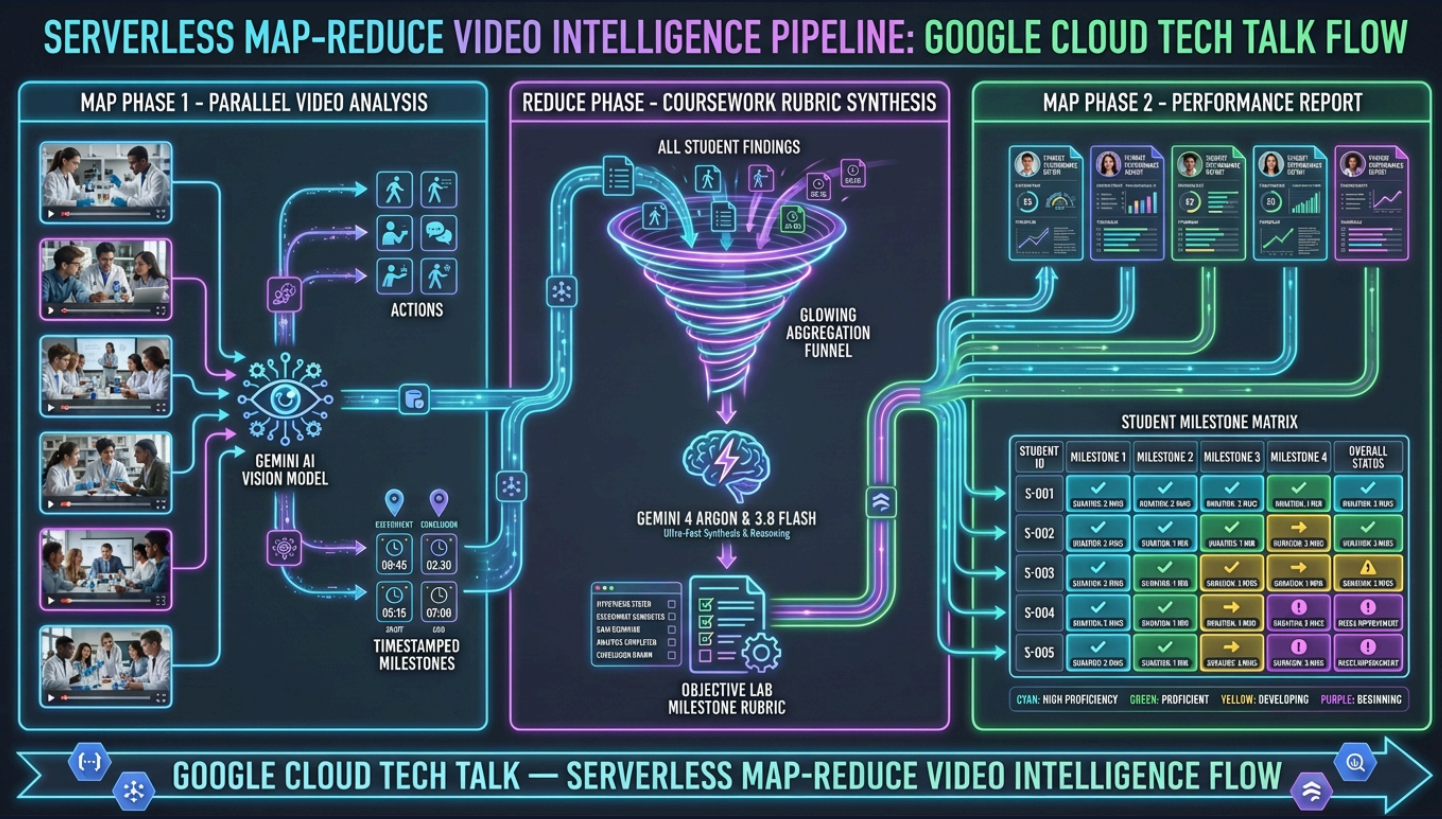
- **Per-Class Dual Retention Policies** (`classes/{classId}`):
  - `retentionDays`: Raw screenshots



07 | Serverless Map-Reduce-Map Video Intelligence Pipeline

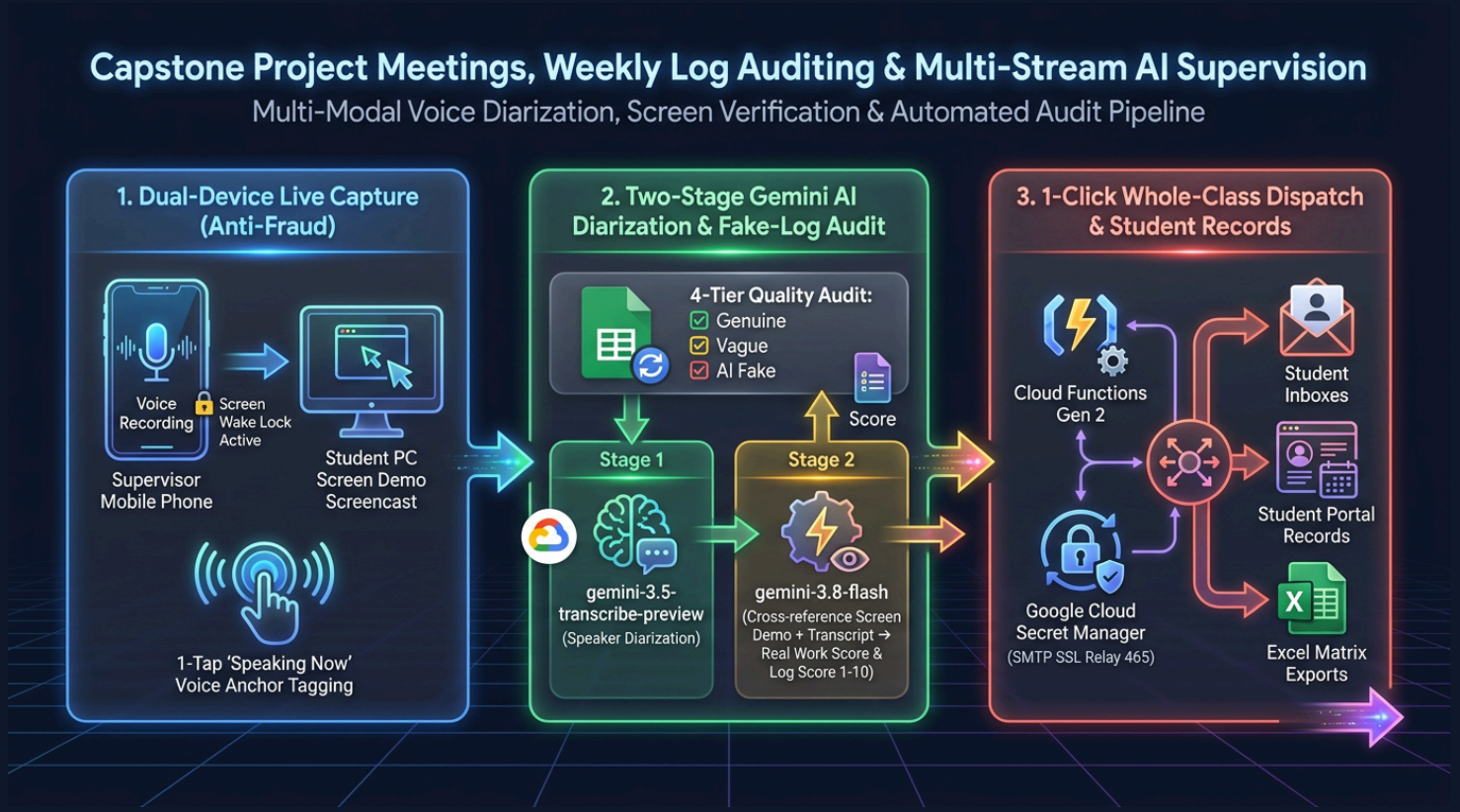
Fan-Out Discovery → Prompt Synthesis → Sortable Milestone Matrix

- **Phase 1: Map (Cloud Tasks Serverless Push Fan-Out):**
  - Master dispatcher initializes progress counters and fans



Capstone Project Meetings & Multi-Stream AI Supervision

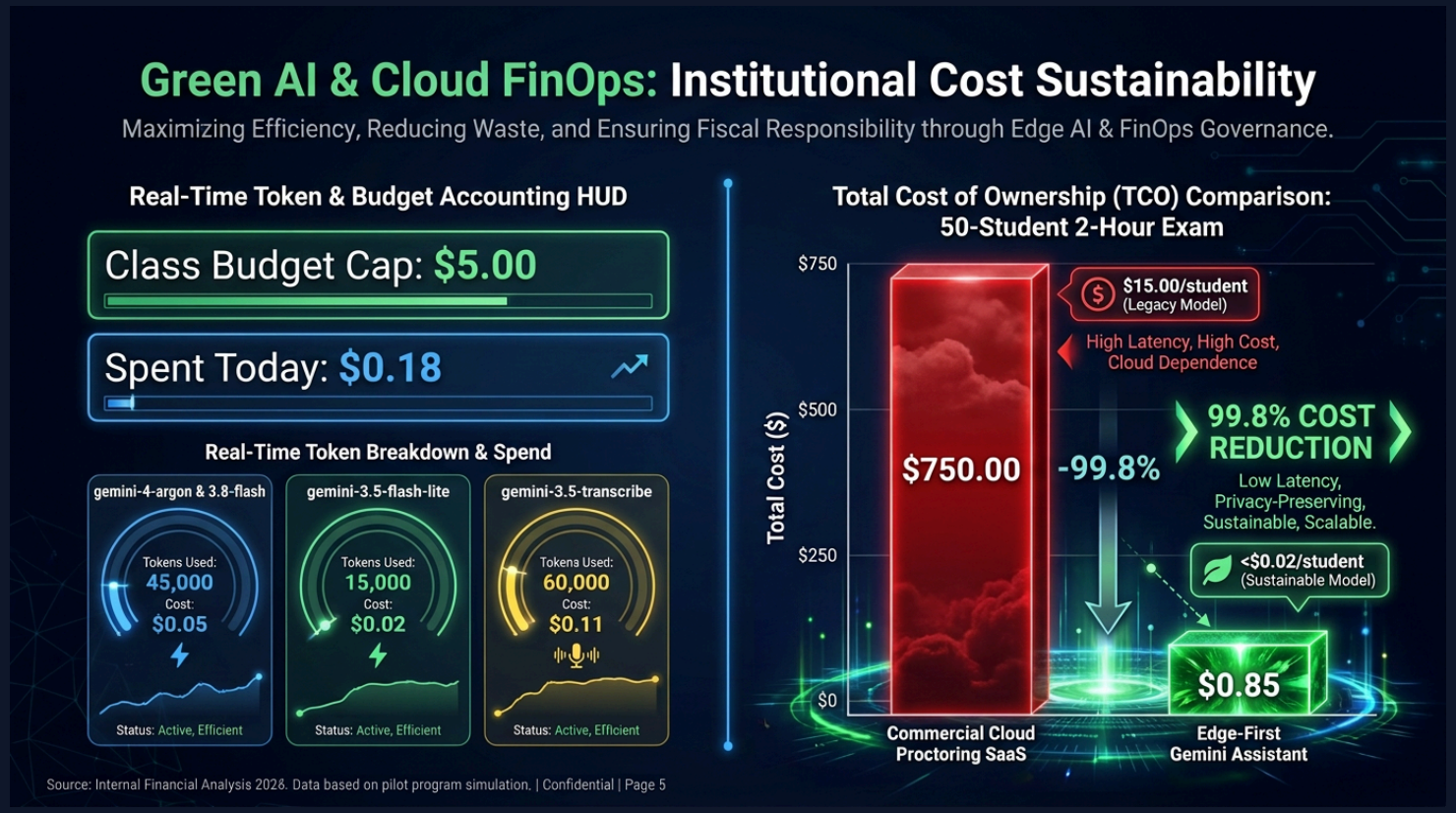
Live Dual-Device Capture, 2-Stage Diarization, Fake-Log AI Audit & Whole-Class Dispatch



- **Dual-Device Live Consultation Capture (Anti-Fabrication):**
  - **Teacher Mobile Voice:** Live

08 | Green AI & Cloud FinOps: Institutional Cost Sustainability

99.8% Cost Reduction: $0.85 per 50-Student Exam vs. $750.00 Commercial SaaS

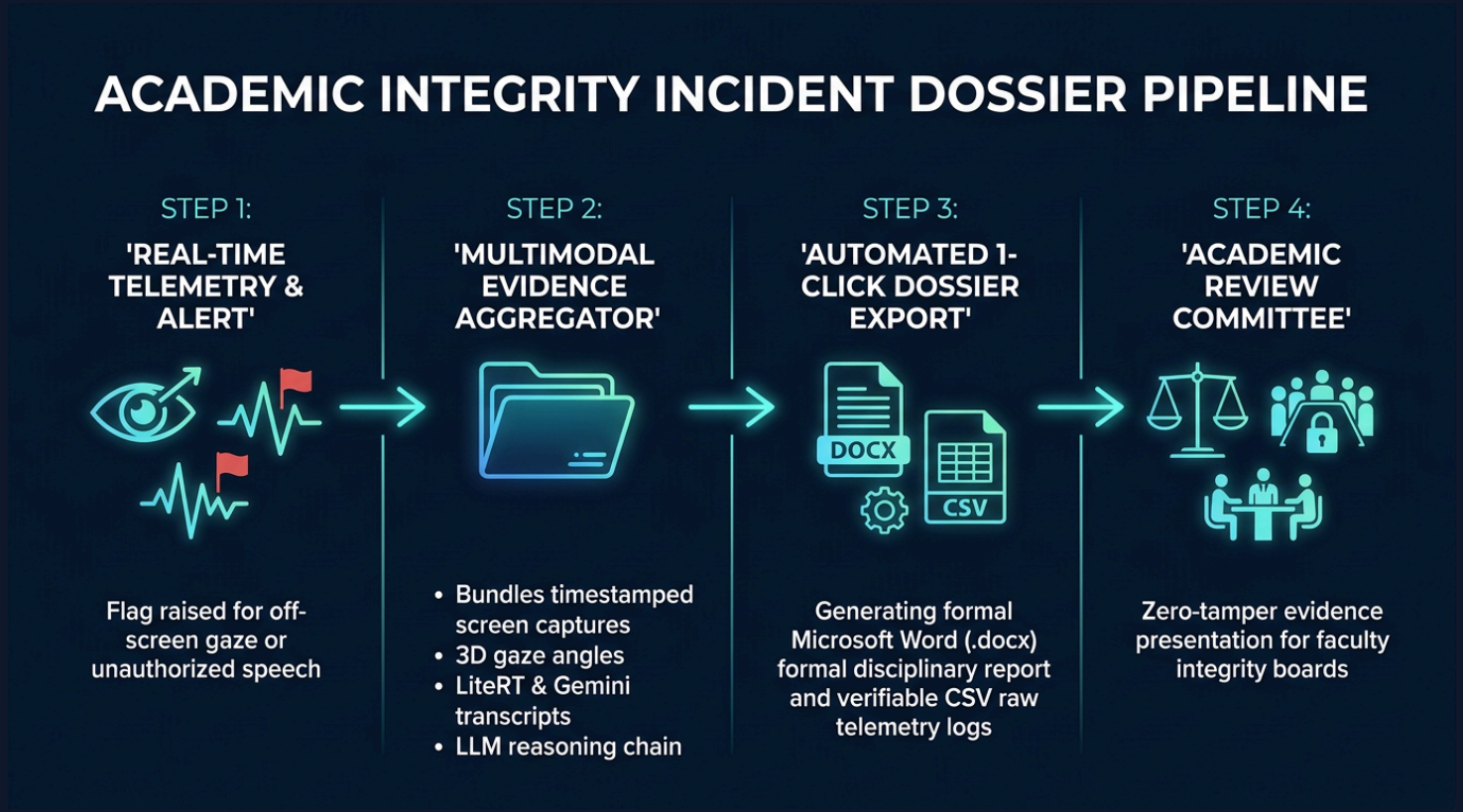


Resource Layer	Commercia Surveillanc SaaS
Compute	100% Cloud

Academic Integrity Incident Dossier Pipeline

Automated 1-Click Export to Verifiable Microsoft Word (.docx) & RFC 4180 CSV

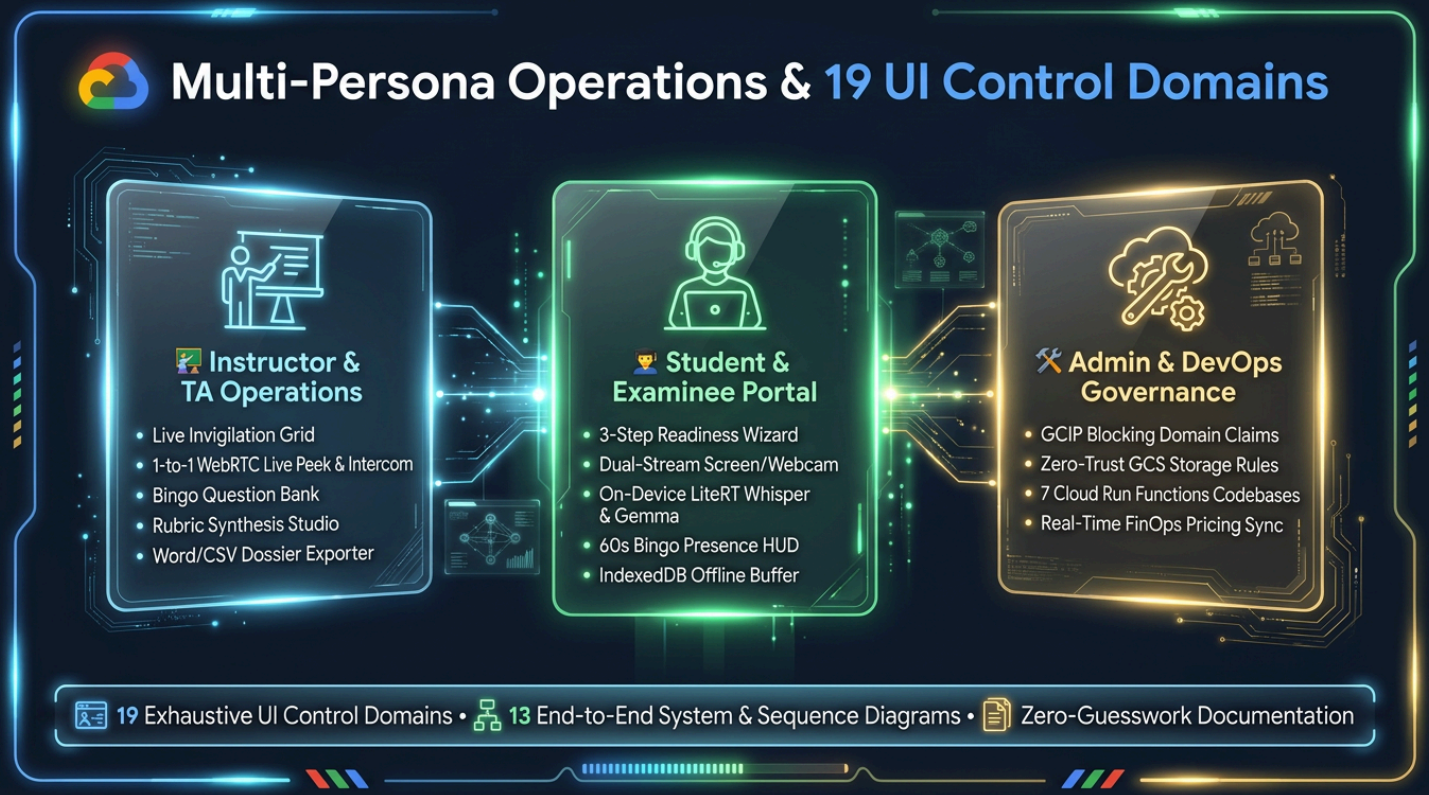
- **Step 1:** Flag raised for gaze diversion, unauthorized speech, or collusion.
- **Step 2:** Multimodal aggregator bundles screenshots, 3D head



Multi-Persona Operations & 19 UI Control Domains

Role-Based Manuals, Student Mobile Portal & Granular UI Control Domains

- **Instructor & TA Command Center (17 Chapters):**
  - Live invigilation grid with zero-space filters, 1-to-1 WebRTC Live



Institutional Admin Console & System Governance Hub

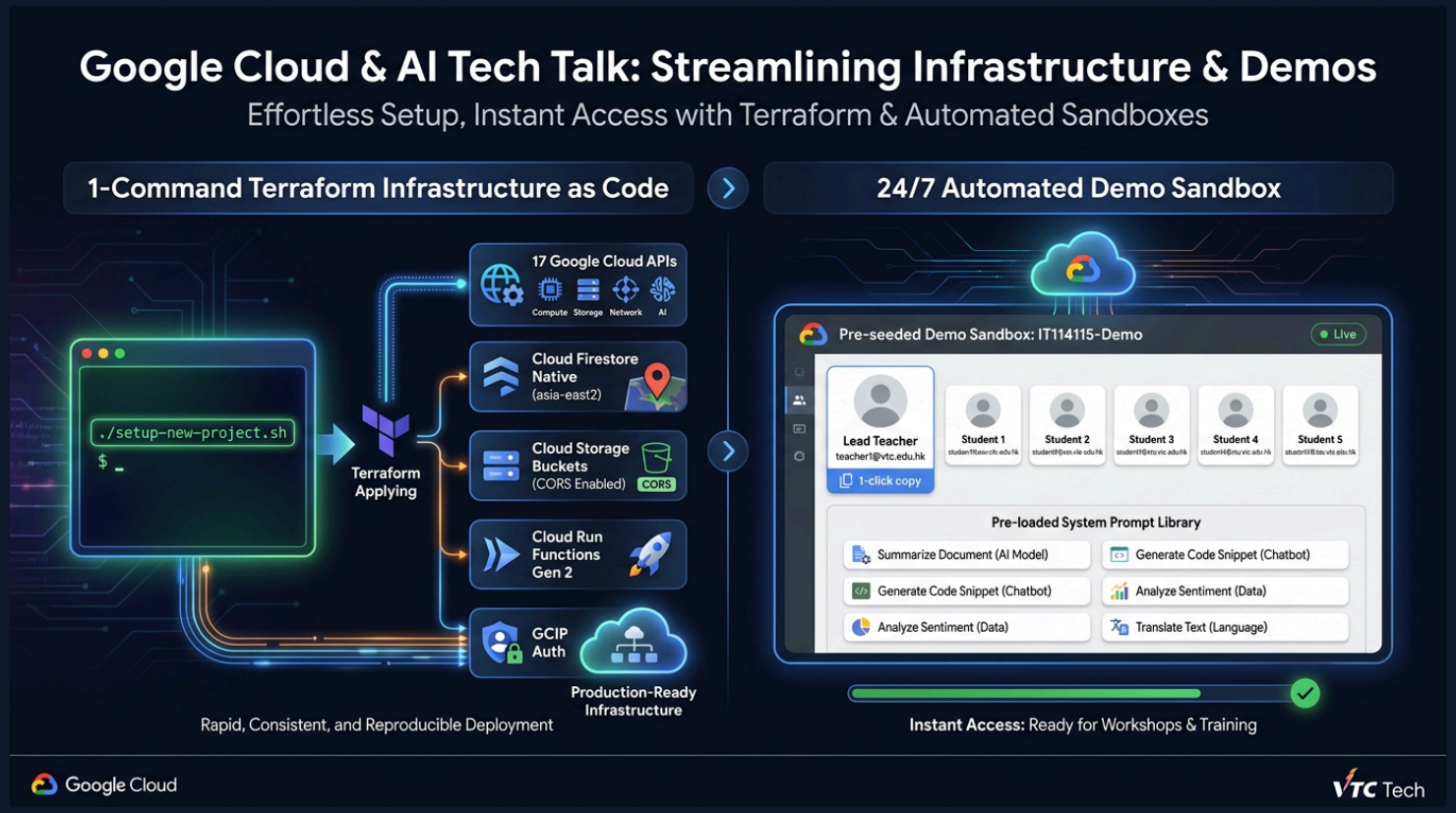
Identity Privilege CLI, Storage Quota Overrides & Automated Prompt Seeding

- **Identity & Tenant Administration (`admin/scripts/`):**
  - GCIP 4-tier role resolution with zero-trust default

1-Command Terraform IaC & Automated Demo Sandbox

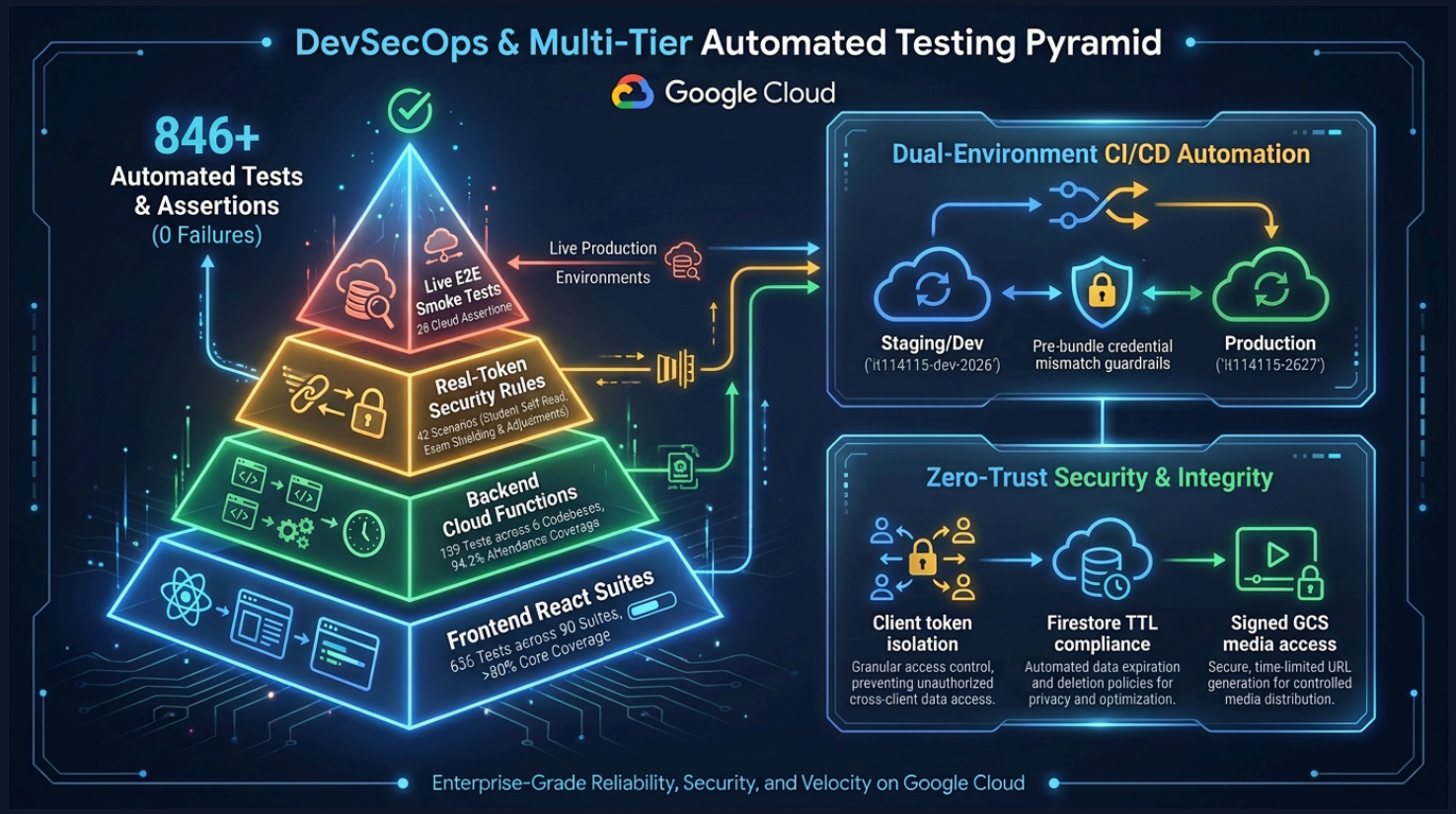
Zero-Click Cloud Provisioning, 24/7 Demo Course & Instant Credential Access

- **1-Command Zero-Click Provisioning (`setup-new-project.sh`):**
  - Single command: `./setup-new-`



09 | DevSecOps & Production Reliability Engineering

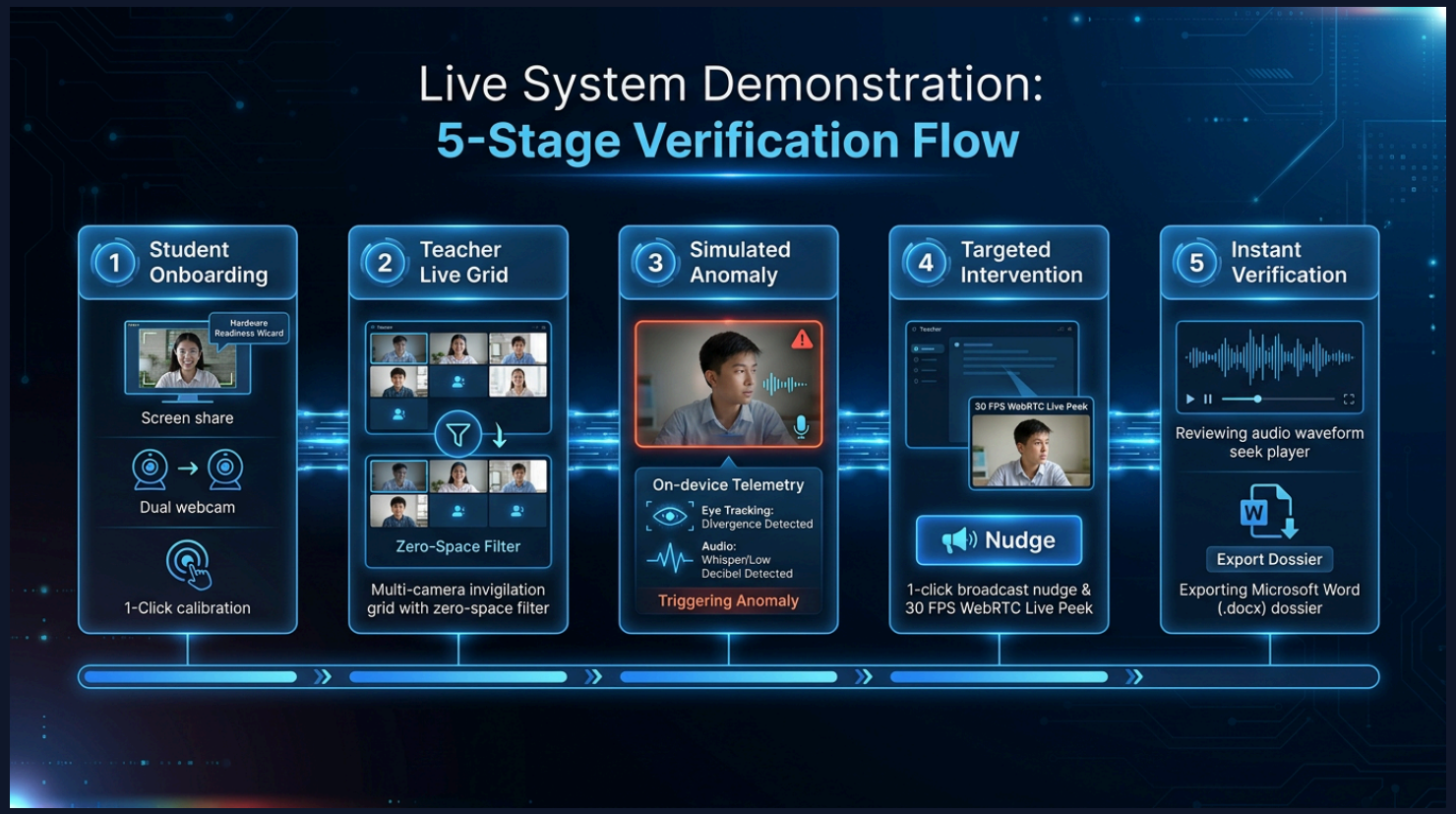
Multi-Tier Automated Testing Pyramid (>2,575 Tests & Assertions) & Gemini 4 & 3 Frontier Standard



- **2,575+ Automated Tests & Assertions (Zero Flaky Tests):**
  - **Level 1 (Frontend):**

Live System Demonstration: 5-Stage Verification Flow

1. **Student Onboarding:** 3-step hardware readiness wizard (Screen share + Dual webcam + 1-Click calibration).
2. **Teacher Live Grid:** Multi-camera invigilation grid with zero-space problem



10 | Empowering Education with Google Cloud & Edge AI

Live System & Open-Source Repository

- **Live Application:** https://it114115-2627.web.app
- **Open-Source Repository:** github.com/wongcyrus/Gemini-AI-Classroom-Assistant
- **Enterprise Documentation:**